

VASP上机

2025年秋季学期

本次上机课内容

- VASP软件，北京超级云计算软件，Linux基本命令介绍
- 以O, H₂, 为例介绍VASP基础计算流程，结构优化...
- 随堂作业
- 课后作业要求

A Simple Molecular Design Strategy for Two-Dimensional Covalent Organic Framework Capable of Visible-Light-Driven Water Splitting

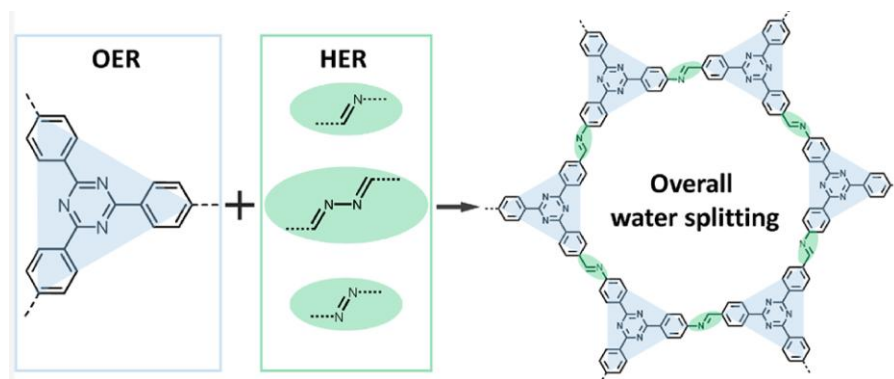
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Cite This: *J. Am. Chem. Soc.* 2020, 142, 4508–4516



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METHODS

Our calculations are performed based on density functional theory (DFT) as implemented in Vienna ab initio simulation package (VASP) with projected augmented wave (PAW) method.^{38–41} The energy cutoffs of 400 and 500 eV for plane wave functions are used for optimization and self-consistent calculations, respectively. A vacuum region of about 15 Å is used to eliminate interaction of layers for 2D nitrogen-linked COFs. Brillouin zones are sampled with $2 \times 2 \times 1$ and $4 \times 4 \times 1$ Γ -centered k -points for L-TA, L-TBZ, L-TST, and L-BZ (L = I, Ai, and Az), respectively. Both lattice constants along the periodic direction and atomic positions are optimized with the convergence of energy and force less than 10^{-5} eV and 0.01 eV/Å, respectively. The screened hybrid HSE06 functional is employed in electronic and band alignment calculations.⁴² BOMD simulations are performed in canonical ensemble with a Nosé thermostat method under the temperature of 300 K for 5 ps with time step of 1 fs. The free energy change (ΔG) during HER and OER on the surface of 2D nitrogen-linked COFs is calculated based on the computational hydrogen electrode (CHE) model developed by Nørskov et al.⁴³ Computational details are available in the [Supporting Information](#).

Single Mo Atom Supported on Defective Boron Nitride Monolayer as an Efficient Electrocatalyst for Nitrogen Fixation: A Computational Study

2. MODELS AND METHODS

All of the computations were performed by means of spin-polarized density functional theory (DFT) methods using the DMol³ code.^{66,67} The Perdew, Burke, and Ernzerhof (PBE)⁶⁸ exchange-correlation functional within a generalized gradient approximation (GGA) was employed (Table S1). The van der Waals interactions were described using the empirical correction in the Grimme scheme,⁶⁹ and the density functional semicore pseudopotential (DSPP) was adopted for the relativistic effects of TM atoms, in which the core electrons are replaced by a single effective potential and some degree of relativistic corrections are introduced into the core.⁷⁰ The double numerical plus polarization (DNP) was chosen as the basis set for other elements, whose accuracy can be comparable to that of Pople's 6-31G** basis set.⁷¹ Self-consistent field (SCF) calculations were performed with a convergence criterion of 10^{-6} au on the total energy and electronic computations. To ensure high quality results, the real-space global orbital cutoff radius was chosen as high as 5.2 Å. Because of the high computational cost, the bulk water layer was not considered because it only slightly stabilizes the NRR intermediates via hydrogen bonding (~ 0.10 eV).⁷² Instead, a conductor-like screening model (COSMO) was used to simulate a H₂O solvent environment,⁷³ whose dielectric constant was set as 78.54. Other solvents, such as THF and Et₂O for NRR, are also used for NRR, which will affect the amount of produced ammonia.⁷⁴

To model the defective BN monolayer with boron monovacancy, we first built a periodic supercell containing 16 boron and 16 nitrogen atoms with a vacuum of 15 Å in the *z*-direction, and then, one boron atom was removed to create a B monovacancy to provide an anchoring site for a single TM atom, in which all of the atoms were fully optimized without any constraints. In order to examine the validity of computations in a 4 × 4 supercell, we recalculated the adsorption energies of some adsorbates on the BN nanosheet in a 5 × 5 supercell. The rather small differences in the computed adsorption energies by using 4 × 4 and 5 × 5 supercells ([Table S2](#)) ensure the reliability of the conclusions in this work. The *k*-point sampling of the Brillouin zone was done using a 5 × 5 × 1 grid centered at the gamma (Γ) point for geometry optimizations, while a 12 × 12 × 1 *k*-points grid was used for electronic structure computations. The Hirshfeld charge analysis was performed to compute the charge transfer and magnetic moment.⁷⁵ The N₂ dissociation minimum energy path (MEP) was obtained by LST/QST tools in DMol³ code.⁷⁶



VASP简介

官网: <https://www.vasp.at/>

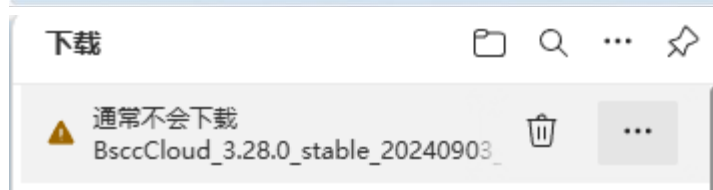
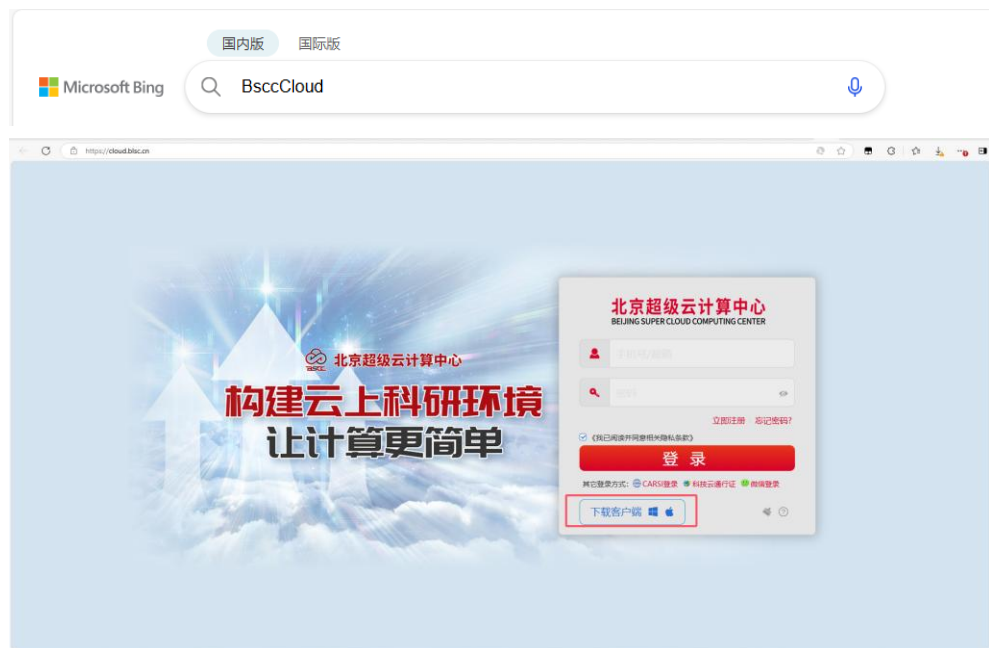
手册: https://vasp.at/wiki/The_VASP_Manual

VASP (Vienna Ab-initio Simulation Package) 是一个广泛使用的量子力学计算软件包, 主要用于基于 密度泛函理论 (Density Functional Theory, DFT) 进行固体物理、材料科学、化学等领域的第一性原理计算。它由奥地利维也纳大学 (University of Vienna) 的计算物理研究小组开发, 广泛应用于电子结构计算、材料性质预测、分子动力学模拟等任务。

- 基于密度泛函理论 (DFT): VASP实现了基于DFT的电子结构计算方法, 采用平面波基组 (plane-wave basis set) 来处理周期性固体体系的电子结构。
- 广泛的功能: VASP不仅可以计算基态的电子结构, 还能够计算能带结构、态密度、光学性质、力学性质、分子动力学 (MD)、反应动力学等。
- 处理复杂体系的能力: VASP能够处理多种复杂的物理化学问题, 包括:
 - 固体、表面、纳米材料和分子系统。
 - 磁性、超导性、能带结构等。
 - 高压、高温条件下的材料性能。

环境

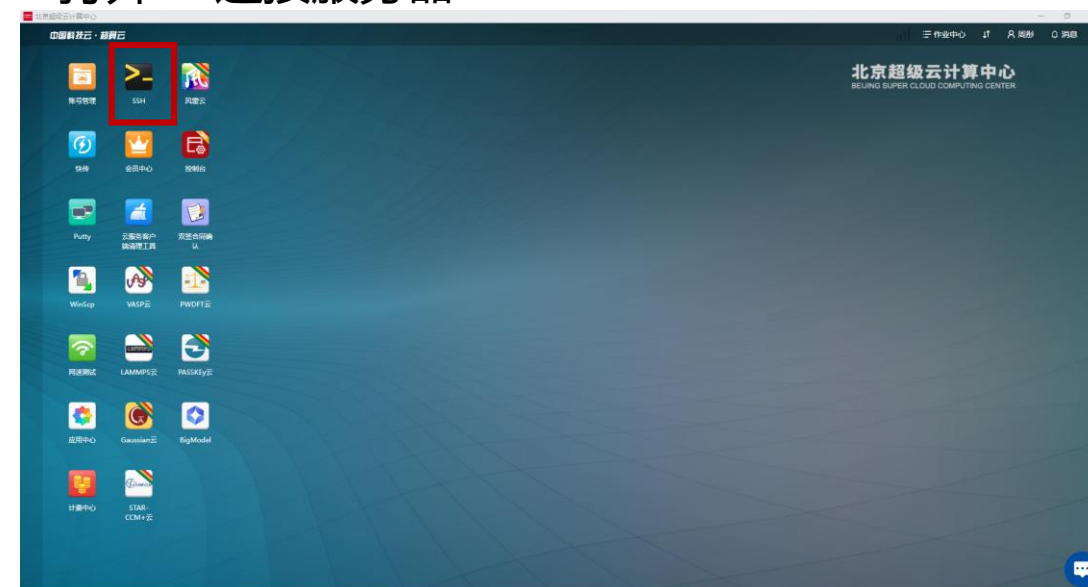
- 登录软件：BssCloud
- 计算用服务器：单节点64核, AMD EPYC 7452
@2.35GHz, 256G内存, DDR4, IB高速网
- 建模软件：Material Studio, VESTA
- 计算软件：VASP&DMol³



<https://cloud.blsc.cn/>
 搜索进入网址
 下载客户端
 可能会显示通常不会下载
 点..., 左键保留
 显示详细信息
 仍然保留



打开ssh连接服务器




```
[a2s000087@ln32%bscc-a2 dft2025]$ ls
Au h2 h2-opt O potpaw_LDA potpaw_PBE pot.sh vasp-pseudopotential.zip vasp.sh
[a2s000087@ln32%bscc-a2 dft2025]$ ls -l
total 54392
drwxrwxr-x  2 a2s000087 a2s000087    4096 Oct 26 23:36 Au
drwxrwxr-x  2 a2s000087 a2s000087    4096 Oct 27 01:18 h2
drwxrwxr-x  2 a2s000087 a2s000087    4096 Oct 27 01:37 h2-opt
drwxrwxr-x  2 a2s000087 a2s000087    4096 Oct 26 22:53 O
drwxrwxr-x 297 a2s000087 a2s000087   20480 Oct 26 19:06 potpaw_LDA
drwxrwxr-x 329 a2s000087 a2s000087   20480 Oct 26 19:06 potpaw_PBE
-rw-r--r--  1 a2s000087 a2s000087     545 Oct 26 19:17 pot.sh
-rw-r--r--  1 a2s000087 a2s000087 55627333 Oct 26 19:09 vasp-pseudopotential.zip
-rwxrwxr-x  1 a2s000087 a2s000087     240 Oct 26 19:34 vasp.sh
[a2s000087@ln32%bscc-a2 dft2025]$ mkdir test
[a2s000087@ln32%bscc-a2 dft2025]$ ls
Au h2 h2-opt O potpaw_LDA potpaw_PBE pot.sh test vasp-pseudopotential.zip vasp.sh
[a2s000087@ln32%bscc-a2 dft2025]$ cd test/
[a2s000087@ln32%bscc-a2 test]$ ls
[a2s000087@ln32%bscc-a2 test]$ vi a.sh
```

ls显示当前文件夹下文件

ls -l 显示详细信息，包括文件大小(单位为字节)，修改日期...

mkdir 创建文件夹

cd 进入文件夹，linux中可用tab键自动补全，比如输入cd te，按tab自动补全为test

vi/vim 进入文本编辑器

创建文件也可以用touch命令，如 touch a.sh



vim文本编辑器：进入后按i进行编辑，键盘上下左右控制光标移动，鼠标左键选中文本可以直接复制，鼠标右键点击可以直接粘贴文本，文本编辑好后，按esc键，然后输入 :wq（注意这里的冒号也要输入）保存并退出，如果修改了不想保存，就输入:q!退出。

按esc，/，然后输入你想搜索的文本，就可以匹配了，按键盘N键，移动到下一个搜索目标

按esc，ctrl+F向前翻页，ctrl+B向后翻页，gg 回到首行，shift+G去到末行，dd删除光标所在行

```
[a2s000087@ln32%bscc-a2 test]$ cd ..
[a2s000087@ln32%bscc-a2 dft2025]$ ls
Au h2 h2-opt 0 potpaw_LDA potpaw_PBE pot.sh test vasp-pseudopotential.zip vasp.sh
[a2s000087@ln32%bscc-a2 dft2025]$ mkdir a
[a2s000087@ln32%bscc-a2 dft2025]$ rm a
rm: cannot remove 'a': Is a directory
[a2s000087@ln32%bscc-a2 dft2025]$ rm -r a
[a2s000087@ln32%bscc-a2 dft2025]$ mkdir a
[a2s000087@ln32%bscc-a2 dft2025]$ ls
a Au h2 h2-opt 0 potpaw_LDA potpaw_PBE pot.sh test vasp-pseudopotential.zip vasp.sh
[a2s000087@ln32%bscc-a2 dft2025]$ mv a b
[a2s000087@ln32%bscc-a2 dft2025]$ ls
Au b h2 h2-opt 0 potpaw_LDA potpaw_PBE pot.sh test vasp-pseudopotential.zip vasp.sh
[a2s000087@ln32%bscc-a2 dft2025]$ cp test/a.sh b/
[a2s000087@ln32%bscc-a2 dft2025]$ cd b/
[a2s000087@ln32%bscc-a2 b]$ ls
a.sh
```

cd .. 回到上一级目录

rm 命令删除文件
这里删除不掉a这个文件夹
因为对文件夹操作要用-r
即 rm -r a

cp 可复制文件

```
[a2s000087@ln32%bscc-a2 b]$ pwd
/public3/home/a2s000087/dft2025/b
[a2s000087@ln32%bscc-a2 b]$ cd ~
[a2s000087@ln32%bscc-a2 ~]$ pwd
/public3/home/a2s000087
[a2s000087@ln32%bscc-a2 ~]$ vim .bashrc
```

pwd显示当前路径

cd ~ 回到家目录

家目录下有一个不可见文件.bashrc
这个文件每次打开服务器都会自动运行，
用来加载环境变量，比如这里加载了
vaspkit, jmol, vtst等

```
1 # .bashrc
2
3 # Source global definitions
4 if [ -f /etc/bashrc ]; then
5     . /etc/bashrc
6 fi
7
8 # Uncomment the following line if you don't like systemctl's auto-paging feature:
9 # export SYSTEMD_PAGER=
10
11 # User specific aliases and functions
12 source /public3/soft/module/module.sh
13 #export PATH=/public3/home/a2s000087/software/vaspkit.1.4.1/bin:${PATH}
14 export PATH=/public3/home/a2s000087/software/vaspkit.1.4.1/bin:${PATH}
15 export PATH=/public3/home/a2s000087/software/jmol-16.2.15:${PATH}
16 export PATH=/public3/home/a2s000087/bin/vtstscripts-1033:${PATH}
17
```

每个小组的组员可在家目录下以
自己的姓名创建一个属于自己的
文件夹，如 mkdir yinxl

组长可创建一个software文件夹用
于公用资源存放

slurm作业管理系统

slurm (Simple Linux Utility for Resource Management) 是一个开源的、高度可扩展的**集群管理和作业调度系统**，专门为高性能计算 (HPC) 环境设计。

- sbatch: 提交批处理作业。
示例: sbatch vasp.sh
- squeue: 查看作业队列状态。
- scancel: 取消作业。
scancel <jobid> # 取消指定作业
- scontrol: 查看和修改作业配置 (通常由管理员使用) 。
scontrol show job <jobid> # 显示作业的详细信息
- sinfo: 查看分区和节点状态。
- scontrol show node: 显示节点的详细信息。

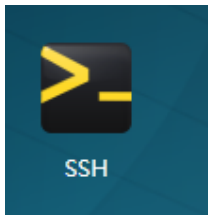
squeue查看任务状态ST:
PD 排队
R 运行
CG 任务完成, 正在退出

每个小组的机时有限, 请大家管理好自己的作业

Commonly Used Linux Commands	
Command	Description
man	display details about an instruction
ls	list contents of the current directory
ls -l	detailed listing of directory contents, shows permissions, owner, etc.
ls -a	list all files (including hidden files)
ls -la	detailed listing of all files (note that options can be combined)
cd	change directory
cd ../	backup one level from the current directory
pwd	print current working directory
touch	create an empty file
mkdir	make a new directory
rm	remove files and directories
rm -rf	recursively remove all files and directories under the specified directory
cat	list file contents
less	list file contents – one screen at a time
tail	list the end of the file (default – displays last 10 lines)
tail -n	list the last n lines of a file
cp	copy file
mv	rename a file
echo	echo values to the screen example: echo \$PATH – prints the value of the PATH variable
grep	command line text search utility example: grep blue colors.txt – list all lines with the word blue from the colorlist.txt file
ps	list currently running processes

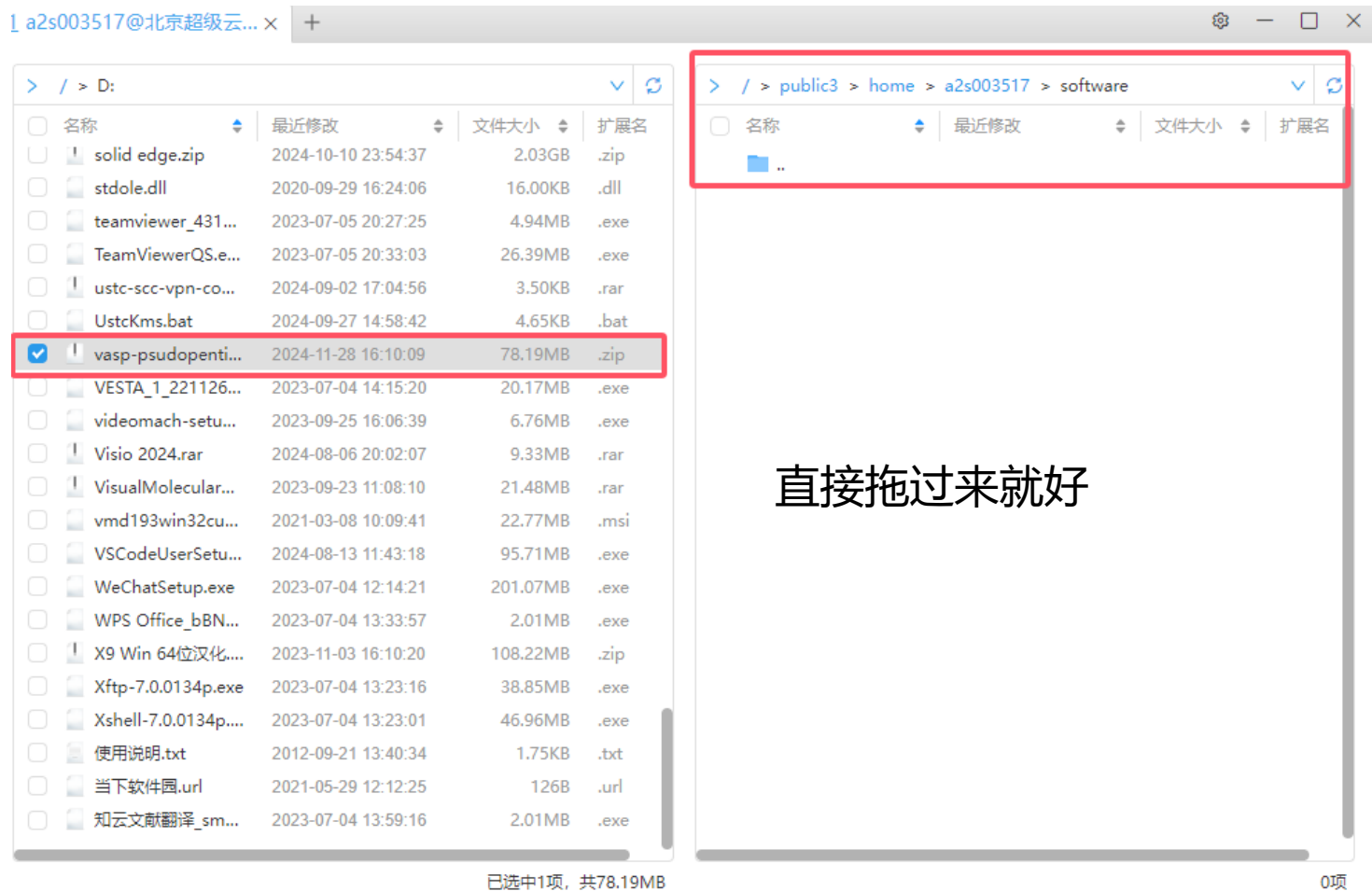
文件和目录管理

- **ls**: 列出当前目录中的文件和子目录。
- **pwd**: 显示当前工作目录的路径。
- **cd**: 切换工作目录。
- **mkdir**: 创建新目录。
- **rmdir**: 删除空目录。
- **rm**: 删除文件或目录。
- **cp**: 复制文件或目录。
- **mv**: 移动或重命名文件或目录。
- **touch**: 创建空文件或更新文件的时间戳。
- **cat**: 连接和显示文件内容。



```
[a2s003517@ln111%bscc-a2 ~]$ ls
pot.sh  software
[a2s003517@ln111%bscc-a2 ~]$
```

- 通过快传，在自己服务器中添加赆势文件和pot.sh, vasp.sh
- 点击SSH,连接
- 输入命令ls, 查看
- 通过unzip xx.zip解压缩
- 进入VASP_PBE, pwd得到目录，这就是PBE的所有赆势
- vi pot.sh,将repo引号中改为PBE赆势的路径后保存 ESC : wq
- 然后可以查看



```
[a2s003517@ln111%bscc-a2 ~]$ cd software/
[a2s003517@ln111%bscc-a2 software]$ ls
vasp-psudopential.zip
[a2s003517@ln111%bscc-a2 software]$ unzip vasp-psudopential.zip
```

解压缩

```
[a2s003517@ln111%bscc-a2 software]$ ls
vasp-psudopential vasp-psudopential.zip
[a2s003517@ln111%bscc-a2 software]$ cd vasp-psudopential/
[a2s003517@ln111%bscc-a2 vasp-psudopential]$ ls
vasp
[a2s003517@ln111%bscc-a2 vasp-psudopential]$ cd vasp/
[a2s003517@ln111%bscc-a2 vasp]$ ls
vasp_LDA vasp_PBE
[a2s003517@ln111%bscc-a2 vasp]$ cd vasp_PBE/
[a2s003517@ln111%bscc-a2 vasp_PBE]$ pwd
/public3/home/a2s003517/software/vasp-psudopential/vasp/vasp_PBE
```

```
#!/bin/bash
# chmod 755 pot    cp pot .    ./pot
repo="/public3/home/a2s003517/software/vasp-psudopential/vasp/vasp_PBE"
sed -i 's/\r//' POSCAR
sed -i '6s/$/& /g' POSCAR
if [ -f POTCAR ]; then
    mv -f POTCAR old-POTCAR
    echo -e "\e[1;31m warning: old POTCAR file found and renamed to 'old-POTCAR' \e[0m "
fi

a=$(sed -n 6p POSCAR)
for i in $a
do
    if test -f $repo/$i/POTCAR ;then
        cat $repo/$i/POTCAR>> POTCAR
    else
        echo -e "\e[1;31m warning:No suitable POTCAR for element ' $i ' found!! skipped this element \e[0m "
    fi
done
sed -n 6p POSCAR && grep TIT POTCAR
~
~
~
~
~
~
```

修改路径

```
[a2s003517@ln111%bsec-a2 ~]$ vi pot.sh
[a2s003517@ln111%bsec-a2 ~]$ grep public pot.sh
repo="/public3/home/a2s003517/software/vasp-psudopential/vasp/vasp_PBE"
[a2s003517@ln111%bsec-a2 ~]$
```



提交脚本，改path

```
+-----+
| 尊敬的用户您好：                                     |
| 各位老师好！bscc-a3系统的应用软件都采用module方式管理，登录系统后可以使用命令module avail查看可以使用 |
| 的软件。为了提高用户使用体验，保障操作流畅性，登陆时不会自动加载module，若有module需求：             |
| public用户可执行source /public1/soft/modules/module.sh                                         |
| 3月14日起，BSCC-A3超算sacct单次查询历史作业详单的最长时间区间调整为30天，查询命令如下：           |
| sacct -D -T -X -u sc**** -S 2022-01-01T00:00:00 -E 2022-01-31T00:00:00                         |
| 温馨提示：                                             |
| 登录节点只能用来进行编辑编译工作，不能运行作业，管理员发现用户在登录节点提交程序，会杀掉相应作业，敬请遵 |
| 守规定！超算是独占节点形式，不满核提交将按照满核收费，建议尽量满核提交作业！                   |
| 超算云服务团队                                       |
| 2022年03月08日                                       |
+-----+
[a2s003517@ln1111%bscc-a2 ~]$
```

```
1 #!/bin/sh
2 #SBATCH -J name 任务名
3 #SBATCH -p amd_256
4 #SBATCH -N 1
5 #SBATCH -n 64 核数
6 source /public3/soft/module/module.sh
7 module load mpi/intel/17.0.5-cjj-public3
8 export PATH=/public3/home/a2s000087/software/vasp.5.4.4/bin:$PATH VASP所在路径
9 mpirun -np 64 vasp_std 运行vasp_std
```

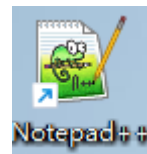

VASP文件

- **INCAR** in **
- STOPCAR in
- stout out
- **POTCAR** in **
- **KPOINTS** in **
- IBZKPT out
- **POSCAR** in **
- CONTCAR out
- **CHG** in/out
- CHG out
- **WAVECAR** in/out
- **TMPCAR** in/out
- EIGENVAL out
- DOSCAR out
- PROCAR out
- OSZICAR out
- PCDAT out
- XDATCAR out
- LOCPOT out
- ELFCAR out
- PROOUT out

VASP 所有的输入文件都只是文本而已 (WAVECAR也可以当成输入文件, 非文本格式无后缀)

杜绝使用Windows自带的文本编辑器: 记事本, 写字板等!!!

Windows下可用Notepad++, linux下用vim



INCAR 告诉VASP算什么, 怎么算。

KPOINTS 包含计算的K点信息(K点是什么, 这个问题很抽象, 脑子里先记住, 计算的时候必须有这个东西)。

POSCAR 是计算模型结构的文字描述, 也就是各个原子在xyz方向的坐标信息 (通过MS等建模软件得到)。

POTCAR 对应的是计算过程中每个原子的赝势, 描述原子中原子核和核电子的相关信息。

```
135437.err CHG CONTCAR EIGENVAL fix.sh INCAR OSZICAR PO POSCAR POTCAR PROCAR vasprun.xml WAVECAR
135437.out CHG CAR DOSCAR f IBZKPT KPOINTS OUTCAR PCDAT POSCAR.cif pot.sh REPORT vaspsh XDATCAR
```

四个必要的输入文件!!!

INCAR

- 精度控制

- PREC、ENCUT、EDIFF、EDIFFG

$$|\mathbf{G} + \mathbf{k}| < G_{\text{cut}} \text{ with } E_{\text{cut}} = \frac{\hbar^2}{2m} G_{\text{cut}}^2$$

- 任务控制

- IBRION, POTIM

- 电子、磁结构控制

- ISPIN、NELECT、NBANDS、LDA+U参数、MAGMOM、NUPDOWN、Dipole

修正

- 展宽

- ISMear, SIGMA

- 输出控制

- LWAVE、LCHARG等

METHODS

Our calculations are performed based on density functional theory (DFT) as implemented in Vienna ab initio simulation package (VASP) with projected augmented wave (PAW) method.^{38–41} The energy cutoffs of 400 and 500 eV for plane wave functions are used for optimization and self-consistent calculations, respectively. A vacuum region of about 15 Å is used to eliminate interaction of layers for 2D nitrogen-linked COFs. Brillouin zones are sampled with $2 \times 2 \times 1$ and $4 \times 4 \times 1$ Γ -centered k -points for L-TA, L-TBZ, L-TST, and L-BZ (L = I, Ai, and Az), respectively. Both lattice constants along the periodic direction and atomic positions are optimized with the convergence of energy and force less than 10^{-5} eV and 0.01 eV/Å, respectively. The screened hybrid HSE06 functional is employed in electronic and band alignment calculations.⁴² BOMD simulations are performed in canonical ensemble with a Nosé thermostat method under the temperature of 300 K for 5 ps with time step of 1 fs. The free energy change (ΔG) during HER and OER on the surface of 2D nitrogen-linked COFs is calculated based on the computational hydrogen electrode (CHE) model developed by Nørskov et al.⁴³ Computational details are available in the [Supporting Information](#).

改变参数来通知计算细节

PREC

Page [Discussion](#) [Read](#) [View source](#) [View history](#)

PREC = Normal | Single | SingleN | Accurate | Low | Medium | High

Default: **PREC** = Medium for VASP4.X
= Normal since VASP5.X

Description: **PREC** specifies the "precision" mode.

PREC sets default values for the energy cutoff **ENCUT**, the FFT grids (**NGX,NGY,NGZ**) and (**NGXF,NGYF,NGZF**), and the accuracy of the projectors in real space **ROPT** (used only when **LREAL**=**TRUE**). Details are given below in the table.

We recommend using **PREC**=Normal or **PREC**=Accurate. **PREC**=Normal can be used for most routine calculations. **PREC**=Accurate leads to a denser grid (**NGX,NGY,NGZ**). Thus, it reduces egg-box effects and strictly avoids any aliasing/wrap-around errors. **PREC**=Normal and **PREC**=Accurate use an augmentation fine grid (**NGXF,NGYF,NGZF**) that is twice larger than the grid (**NGX,NGY,NGZ**) used for the representation of the pseudo-orbitals. **PREC**=Accurate increases the memory requirements somewhat, but it should be used (in combination with an increased value for **ENCUT**) when a very good accuracy is required, e.g., for accurate forces, for phonons and stress tensor or in general when second derivatives are computed. The accuracy of forces can also be sometimes further improved by specifying **ADDGRID**=**TRUE**, however, reports from users are somewhat contradictory about whether this really helps.

NGX, NGY, NGZ	FFT mesh for orbitals (Sec. 6.3 , 6.11)
NGXF,NGYF,NGZF	FFT mesh for charges (Sec. 6.3 , 6.11)
<i>NBANDS</i>	number of bands included in the calculation (Sec. 6.5)
NBLK	blocking for some BLAS calls (Sec. 6.6)
<i>SYSTEM</i>	name of System
NWRITE	verbosity write-flag (how much is written)
ISTART	startjob: 0-new 1-cont 2-samecut
ICHARG	charge: 1-file 2-atom 10-const
ISPIN	spin polarized calculation (2=yes 1-no)
MAGMOM	initial mag moment / atom
INIWAV	initial electr wf. : 0-lowe 1-rand
<i>ENCUT</i>	energy cutoff in eV

<i>PREC</i>	precession: medium, high or low
<i>PREC</i>	VASP.4.5 also: normal, accurate
NELM, NELMIN and NELMDL	nr. of electronic steps
EDIFF	stopping-criterion for electronic upd.
EDIFFG	stopping-criterion for ionic upd.
NSW	number of steps for ionic upd.
NBLOCK and KBLOCK	inner block; outer block
IBRION	ionic relaxation: 0-MD 1-quasi-New 2-CG
ISIF	calculate stress and what to relax
IWAVPR	prediction of wf.: 0-non 1-charg 2-wave 3-comb
ISYM	symmetry: 0-nonsym 1-usesym
SYMPREC	precession in symmetry routines
LCORR	Harris-correction to forces
POTIM	time-step for ion-motion (fs)

电子步最大迭代步数

离子步最大迭代步数

TEBEG, TEEND	temperature during run
SMASS	Nose mass-parameter (am)
NPACO and APACO	distance and nr. of slots for P.C.
POMASS	mass of ions in am
ZVAL	ionic valence
<i>RWIGS</i>	Wigner-Seitz radii
NELECT	total number of electrons
NUPDOWN	fix spin moment to specified value
EMIN, EMAX	energy-range for DOSCAR file
<i>ISMEAR</i>	part. occupancies: -5 Blöchl -4-tet -1-fermi 0-gaus >0 MP
<i>SIGMA</i>	broadening in eV -4-tet -1-fermi 0-gaus
<i>ALGO</i>	algorithm: Normal (Davidson) Fast Very_Fast (RMM-DIIS)
<i>IALGO</i>	algorithm: use only 8 (CG) or 48 (RMM-DIIS)

<i>ROPT</i>	number of grid points for non-local proj in real space
GGA	xc-type: e.g. PE AM or 91
VOSKOWN	use Vosko, Wilk, Nusair interpolation
DIPOL	center of cell for dipol
AMIX, BMIX	tags for mixing
WEIMIN, EBREAK, DEPER	special control tags
TIME	special control tag
<i>LWAVE, LCHARG, LVTOT, LVHAR</i>	create WAVECAR/CHGCAR/LOCPOT
LELF	create ELFCAR
LORBIT	create PROOUT
NPAR	parallelization over bands
LSCALAPACK	switch off scaLAPACK
LSCALU	switch of LU decomposition

DOS计算

O 原子计算: INCAR

- SYSTEM = O atom # 氧符号, 不是零!
- ISMEAR = 0 # 是零, 不是氧符号!
- SIGMA = 0.01

熟知各个参数的数值意义, 以及在不同计算任务中的设置至关重要。

- 每个参数后面跟着等号 = ,
- 等号前后可以有空格, 也可以没有, ISMEAR=0 和 ISMEAR = 0 结果一样,
- SYSTEM #后面是该计算的注释, 这个可以随便写, 可以删除。
- ISMEAR这一项, 不同的值对应的是不同的展宽方法, 对于半导体和绝缘体体系, ISMEAR的值取**绝对不能大于0**, 一般用0, 金属体系一般用1, 更加精确的时候用 -5。
- SIGMA该项, 如果用了 ISMEAR = -5 , 可以忽略; 对于金属: ISMEAR = 1 或者0, 非金属: ISMEAR = 0 的时候, 一般取0.10 或者采用默认值 0.20 即可。SIGMA 的取值要保证OUTCAR 中的 entropy T*S 这一项, 平均到每个原子上小于 1meV。可使用: `grep 'entropy T' OUTCAR`

对于分子, 原子体系, 必须要用很小的值, 这里我们用的0.01。

O 原子计算: KPOINTS

```
K-POINTS # 第一行随便写都行
0        # 零, 格子自动生成
Gamma    # gamma 点 centered, Monkhorst
1 1 1    # 1*1*1格子
0 0 0    # S1 S2 S3, 一般保持 0 0 0 不变。
```

- 1) 第二行是数字0, 不是字母O, 0 在这里表示的是自动生成K点。
- 2) 第三行, VASP只认第一个字母, 大小写均可。另外一种原始的Monkhorst-Pack 网格,
- 3) 第四行, 在xyz三个方向上生成对应数目的K点, 本例是1*1*1;
- 4) 第五行, K点网格位移, 一般都写成 0 0 0 ;
- 5) **对于六角晶系hexagonal的结构, 采用gama centered!** 这一点, vasp官网说的很明确

KPOINTS的选取

- 方法1: **使用官网建议取值**, 每个晶格矢量的长度(a,b,c)乘以这个方向上的K点数目(ka)。需要做收敛性测试
- 比如我要算一个绝缘体体系, a,b的晶胞长度为6Å, c方向真空层15Å, K点就是3 3 1

Hint

A rule of thumb for choosing the initial **k**-point sampling is, that the product, ka , between the number of **k**-points, k , in any direction, and the length of the basis vector in this direction, a , should be:

- $ka \sim 30 \text{ \AA}$, for d band metals
- $ka \sim 25 \text{ \AA}$, for simple metals
- $ka \sim 20 \text{ \AA}$, for semiconductors
- $ka \sim 15 \text{ \AA}$, for insulators

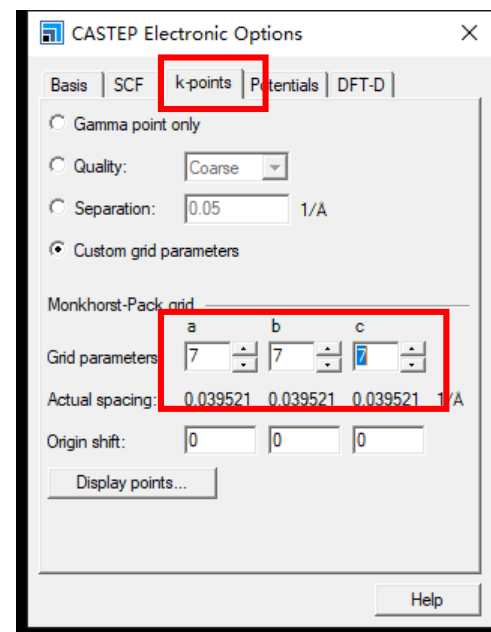
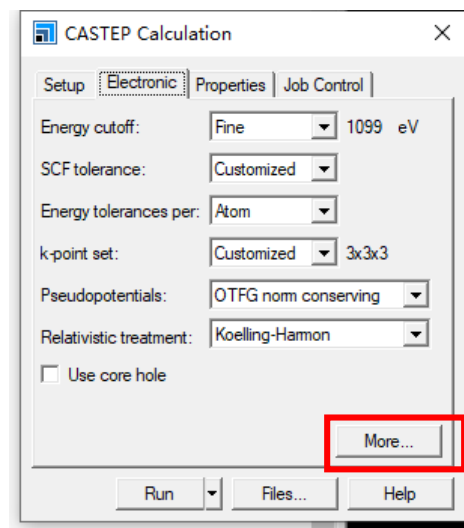
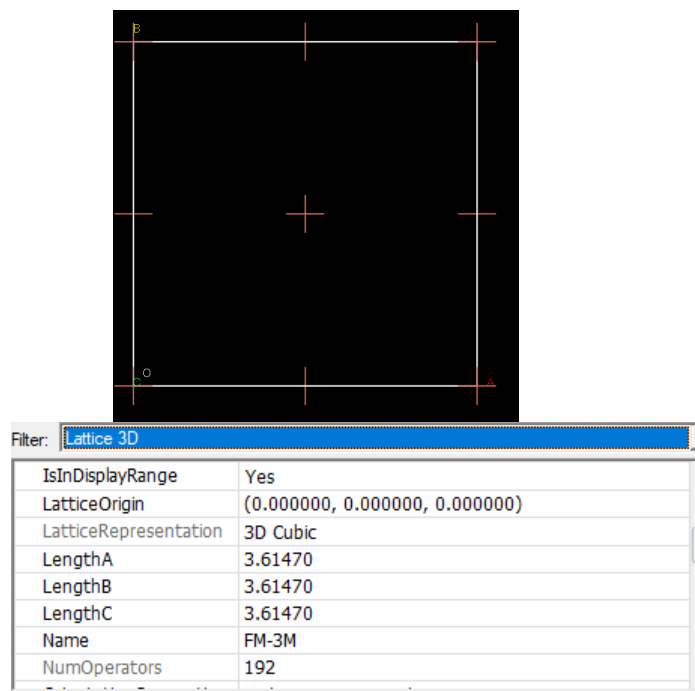
d区金属, $ka \sim 30 \text{ \AA}$
普通金属, $ka \sim 25 \text{ \AA}$
半导体, $ka \sim 20 \text{ \AA}$
绝缘体, $ka \sim 15 \text{ \AA}$

Remember that convergence in this parameter should always be checked.

KPOINTS的选取



- 方法二：MS建模后，点击CASTEP Tools图标，查看不同K点下，K点密度，一般密度在0.03最佳
- 例如，FCC Cu 的KPOINTS 则是 7 7 7



O 原子计算: POSCAR

O atom in a box #1

1.0 ! universal scaling parameters #2

8.0 0.0 0.0 ! lattice vector a(1) #3

0.0 8.0 0.0 ! lattice vector a(2) #4

0.0 0.0 8.0 ! lattice vector a(3) #5

O ! O element #6 not zero 有几种原子写几个

1 ! number of atoms #7

cart ! positions in cartesian coordinates #8

0 0 0 #9

H O

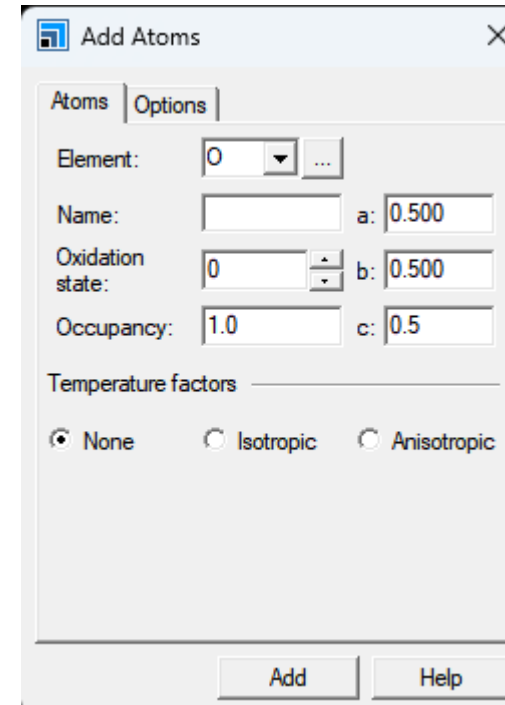
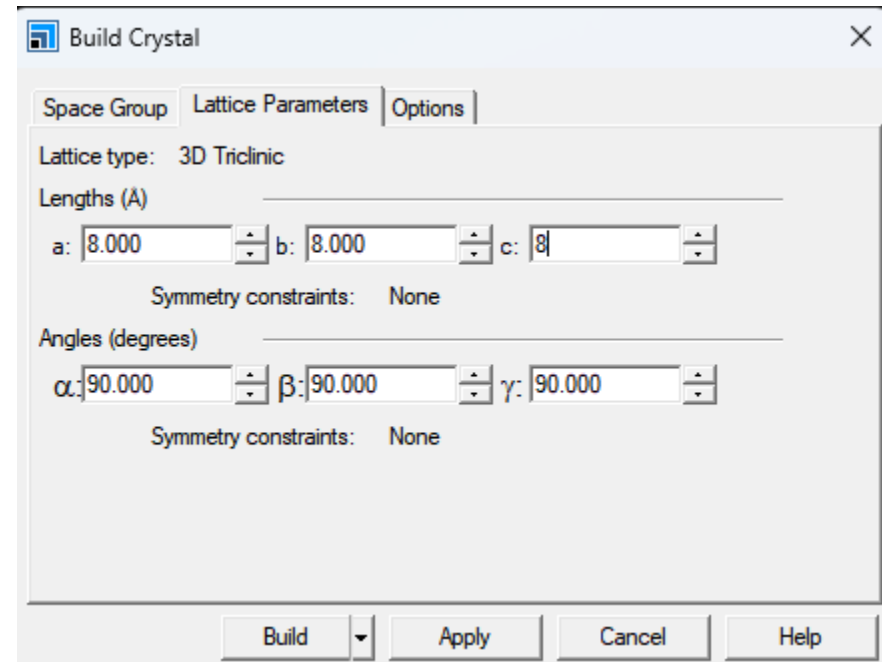
2 1

Selective Dynamics

Cart, Direct

0.0 0.0 0.0 F F F

1.0 0.0 0.0 T T T



MS中建模，导出为.cif文件，然后用VESTA转换为POSCAR文件

.cif结构文件除了用VESTA转换，也可以直接由vaspkit产生POSCAR，具体操作输入：vaspkit → 105

文件名(N):	POSCAR
保存类型(T):	Crystallographic Information Files (*.cif;*.cmf)

```
3D\Atomistic\ (2)
1.0
      8.0000000000      0.0000000000      0.0000000000
      0.0000000000      8.0000000000      0.0000000000
      0.0000000000      0.0000000000      8.0000000000
O
1
Direct
0.5000000000      0.5000000000      0.5000000000
```

文件名(N):	POSCAR
保存类型(T):	VASP (POSCAR;*.vasp)

POTCAR

```
PAW_PBE 0 08Apr2002
6.000000000000000
parameters from PSCTR are:
VRHFIN =0: s2p4
LEXCH = PE
EATOM = 432.3788 eV, 31.7789 Ry

TITEL = PAW_PBE 0 08Apr2002
LULTRA = F use ultrasoft PP ?
IUNSCR = 1 unscreen: 0-lin 1-nonlin 2-no
RPACOR = 1.200 partial core radius
POMASS = 16.000; ZVAL = 6.000 mass and valenz
RCORE = 1.520 outmost cutoff radius
RWIGS = 1.550; RWIGS = 0.820 wigner-seitz radius (au A)
ENMAX = 400.000; ENMIN = 300.000 eV
ICORE = 2 local potential
LCOR = T correct aug charges
LPAW = T paw PP
EAUG = 605.392
DEXC = 0.000
RMAX = 1.553 core radius for proj-oper
RAUG = 1.300 factor for augmentation sphere
RDEP = 1.550 radius for radial grids
RDEPT = 1.329 core radius for aug-charge
```

```
[a2s000087@ln111%bscc-a2 assistant]$ ls
o pot.sh test
[a2s000087@ln111%bscc-a2 assistant]$ cp pot.sh o/
[a2s000087@ln111%bscc-a2 assistant]$ cd o/
[a2s000087@ln111%bscc-a2 o]$ ls
INCAR KPOINTS POSCAR pot.sh
[a2s000087@ln111%bscc-a2 o]$ sh pot.sh
O
TITEL = PAW_PBE 0 08Apr2002
[a2s000087@ln111%bscc-a2 o]$
```

体系中有多种元素时：使用cat命令

cat C H O N >POTCAR

切记和POSCAR中元素顺序保持一致!!!

- VRHFIN 用来看元素的价电子排布,
- LEXCH 表示这个POTCAR对应的是GGA-PBE泛函; 如果INCAR中不设定泛函, 则默认通过这个参数来设定。
- ZVAL 指的是实际上POTCAR中价电子的数目, 做Bader电荷分析的时候, 极其重要。
- ENMAX 代表默认的截断能。若INCAR中不设置ENCUT, 则默认为这个值
- TITEL 指的是哪个元素, 以及POTCAR发布的时间

查看POTCAR有哪些元素 grep TIT POTCAR

```
[a2s000087@ln32%bscc-a2 o]$ grep TIT POTCAR
TITEL = PAW_PBE 0 08Apr2002
```

POTCAR的选取

- VASP提供的赝势文件，并不是每个元素只有一种，打开赝势库看一下，几乎所有元素的赝势都不止包含一种，VASP在手册上有建议使用的赝势

_h, _s	hard, soft
_pv, _sv, _d	包含更内层的电子
_2, _3	含不同价电子数（离子）
_GW	GW计算用
H1.25, H1.5, H.5	赝氢

- **标准赝势没有后缀（最常用）**
- _h赝势是更硬的赝势，(更精确，需要更高截断能)
- _s赝势是更软的赝势，(更粗糙，需要的截断能小)
- _pv赝势是对于d区过渡金属，包含内层的p轨道，比如V_pv的赝势包含3p3d4s,
- _sv赝势是对于d区过渡金属，包含内层的s和p轨道，比如V_sv的赝势包含3s3p3d4s,
- _d是对于主族4周期以后的元素，包含内层d轨道，比如Ga_d的赝势包含3d4s4p

将任务提交脚本vasp.sh也复制进来，就准备好计算的所有文件了

```
[a2s000087@ln111%bscc-a2 o]$ ls
INCAR KPOINTS POSCAR POTCAR pot.sh vasp.sh
[a2s000087@ln111%bscc-a2 o]$ ls
INCAR KPOINTS POSCAR POTCAR pot.sh vasp.sh
[a2s000087@ln111%bscc-a2 o]$ sbatch vasp.sh
Submitted batch job 4981449
[a2s000087@ln111%bscc-a2 o]$ ls
INCAR KPOINTS POSCAR POTCAR pot.sh vasp.sh
[a2s000087@ln111%bscc-a2 o]$ squeue
```

sbatch提交任务

squeue查看任务队列

ls	JOBID	PARTITION	NAME	USER	ST	TIME	NODES	ODELIST (REASON)
	4981449	amd_256	name a2s00008	PD		0:00	1	(Priority)

任务状态ST:
PD 排队
R 运行
CG 任务完成，正在退出

```
[a2s000087@ln111%bscc-a2 o]$ squeue
JOBID PARTITION NAME USER ST TIME NODES ODELIST (REASON)
4981449 amd_256 name a2s00008 R 0:04 1 x1501
[a2s000087@ln111%bscc-a2 o]$ ls
CHG CHGCAR CONTCAR DOSCAR EIGENVAL IBZKPT INCAR KPOINTS OSZICAR OUTCAR PCDAT POSCAR POTCAR pot.sh REPORT slurm-4981449.out vasprun.xml vasp.sh WAVECAR XDATCAR
[a2s000087@ln111%bscc-a2 o]$ squeue
JOBID PARTITION NAME USER ST TIME NODES ODELIST (REASON)
[a2s000087@ln111%bscc-a2 o]$
```

查看作业结果

- **检查任务是否算完有很多方式：**查看队列，查看输出文件。squeue 查看队列
- `tail -f OUTCAR` 可实时监看任务运行状态，`ctrl + C` 退出
- 任务完成后，`tail OUTCAR`（OUTCAR是vasp的主要输出文件），这个命令用以查看文件最后面的几行，如果你的OUTCAR和图中的一样，说明任务已经完成。

```
[a2s000087@ln32%bscc-a2 0]$ tail OUTCAR
      User time (sec):           1.670
      System time (sec):         1.751
      Elapsed time (sec):        3.464

      Maximum memory used (kb):   72740.
      Average memory used (kb):    0.

      Minor page faults:         14720
      Major page faults:          0
      Voluntary context switches: 2451
```

OSZICAR

```
[a2s000087@ln32%bscc-a2 O]$ cat OSZICAR -n
```

		N	E	dE	d eps	ncg	rms	rms (c)
2	DAV:	1	0.270758870670E+02	0.27076E+02	-0.15315E+03	128	0.394E+01	
3	DAV:	2	0.307822345719E-01	-0.27045E+02	-0.27045E+02	192	0.139E+01	
4	DAV:	3	-0.123476546144E-01	-0.43130E-01	-0.37183E-01	128	0.770E-01	
5	DAV:	4	-0.123957341297E-01	-0.48080E-04	-0.48075E-04	256	0.295E-02	
6	DAV:	5	-0.123957354501E-01	-0.13204E-08	-0.90701E-10	128	0.408E-05	0.307E-01
7	DAV:	6	-0.134137852737E-01	-0.10180E-02	-0.17905E-03	128	0.415E-02	0.155E-01
8	DAV:	7	-0.135116082868E-01	-0.97823E-04	-0.23168E-04	192	0.175E-02	
9	1	F=	-.13511608E-01	E0=	-.11022548E-01	d E	==	-.746718E-02

- **N** 代表电子步的迭代步数，本例中里面有7个电子步；
- **E** 代表当前电子步的体系能量；
- **dE** 该步和上一步体系能量的差值；
- **d eps** the change in the band structure energy; 本征值的变化
- **ncg** the number of evaluations of the Hamiltonian acting onto a wavefunction; 波函数的优化次数
- **rms** the norm of the residuum of the trial wavefunctions (i.e. their approximate error)
- **rms(c)** the difference between input and output charge density.
- **DAV**: blocked Davidson iteration scheme 的缩写, 一个电子自洽算法的缩写, RMM (residual minimization scheme) 和 CG (conjugate-gradient algorithm) 等, 这些是由INCAR中的ALGO 参数决定。
- **E0**: DFT总能
- **F**: 自由能 $F=E0-TS$

OUTCAR

vi OUTCAR查看

```
vasp.5.4.4.18Apr17-6-g9f103f2a35 (build Oct 13 2023 10:54:58) complex
executed on           LinuxIFC date 2025.10.26  21:26:23
running on    64 total cores
distrk:  each k-point on    64 cores,    1 groups
distr:  one band on Ncores_PER_BAND=    1 cores,    64 groups
-----
```

VASP会列出来其版本，时间，以及服务器的相关情况

```
INCAR:
POTCAR:  PAW_PBE O 08Apr2002
-----
|
|      W      W      AA      RRRRR      N      N      II      N      N      GGGG      !!!
|      W      W      A      A      R      R      NN      N      II      NN      N      G      G      !!!
|      W      W      A      A      R      R      N      N      N      II      N      N      N      G      !!!
|      W      WW      W      AAAAAA      RRRRR      N      N      N      II      N      N      N      G      GGG      !
|      WW      WW      A      A      R      R      N      NN      II      N      NN      G      G
|      W      W      A      A      R      R      N      N      II      N      N      GGGG      !!!
|
|  For optimal performance we recommend to set
|  NCORE= 4 - approx Sqrt( number of cores)
|  NCORE specifies how many cores store one orbital (NPAR=cpu/NCORE).
|  This setting can greatly improve the performance of VASP for DFT.
|  The default,  NCORE=1  might be grossly inefficient
|  on modern multi-core architectures or massively parallel machines.
|  Do your own testing !!!!
|  Unfortunately you need to use the default for GW and RPA calculations.
|  (for HF NCORE is supported but not extensively tested yet)
|
|-----
```

辨别WARNING给出的信息

下面是 POTCAR 的基本信息，如果你想通过 OUTCAR 查看 POTCAR 中的元素时，可以使用下面的命令：

```
grep POTCAR OUTCAR
```

```
grep TIT OUTCAR
```

```
grep ENMAX OUTCAR
```

.....

ZVAL 是该 POTCAR 中对应元素的价电子，这里氧原子含有6个外层价电子。

```
POTCAR:      PAW_PBE O 08Apr2002
VRHFIN =O: s2p4
LEXCH  = PE
EATOM  =    432.3788 eV,    31.7789 Ry

TITEL  = PAW_PBE O 08Apr2002
LULTRA =          F      use ultrasoft PP ?
IUNSCR =          1      unscreen: 0-lin 1-nonlin 2-no
RPACOR =    1.200      partial core radius
POMASS =    16.000; ZVAL  =    6.000      mass and valenz
RCORE  =    1.520      outmost cutoff radius
RWIGS  =    1.550; RWIGS =    0.820      wigner-seitz radius (au A)
ENMAX  =   400.000; ENMIN =   300.000 eV
ICORE  =          2      local potential
LCOR   =          T      correct aug charges
LPAW   =          T      paw PP
EAUG   =   605.392
DEXC   =    0.000
RMAX   =    1.553      core radius for proj-oper
RAUG   =    1.300      factor for augmentation sphere
RDEP   =    1.550      radius for radial grids
RDEPT  =    1.329      core radius for aug-charge
```

POSCAR 的一些基本信息： 坐标格式， 原子位置， 以及晶胞的形状大小。

```
POSCAR: 3D\Atomistic\ (2)
positions in direct lattice
No initial velocities read in
exchange correlation table for LEXCH =      8
RHO(1)=      0.500      N(1) =      2000
RHO(2)=    100.500      N(2) =      4000
```

```
ion position          nearest neighbor table
  1  0.500  0.500  0.500-

LATTYP: Found a simple cubic cell.
ALAT      =      8.0000000000

Lattice vectors:

A1 = (    8.0000000000,    0.0000000000,    0.0000000000)
A2 = (    0.0000000000,    8.0000000000,    0.0000000000)
A3 = (    0.0000000000,    0.0000000000,    8.0000000000)
```

体系的对称性以及点群操作相关的信息，在这里我们的体系是立方体，为 O_h 的点群，有48个对称操作。

```
769 irot : 48
770 -----
771 isymop: 0 0 1
772         0 1 0
773         1 0 0
774
775 gtrans: 0.0000000 0.0000000 0.0000000
776
777 ptrans: 0.0000000 0.0000000 0.0000000
778 rotmap:
779 ( 1-> 1)
```

K点信息：想查看K点个数： `grep irreducible OUTCAR`

```
[a2s000087@ln32%bscc-a2 0]$ grep irreducible OUTCAR
Found      1 irreducible k-points:
```

```
Subroutine IBZKPT returns following result:
```

```
=====
```

```
Found      1 irreducible k-points:
```

```
Following reciprocal coordinates:
```

Coordinates	Weight
0.000000 0.000000 0.000000	1.000000

```
Following cartesian coordinates:
```

Coordinates	Weight
0.000000 0.000000 0.000000	1.000000

计算参数初始化

Dimension of arrays:

k-points	NKPTS =	1	k-points in BZ	NKDIM =	1	number of bands	NBANDS=	64
number of dos	NEDOS =	301	number of ions	NIONS =	1			
non local maximal	LDIM =	4	non local SUM 2l+1	LMDIM =	8			
total plane-waves	NPLWV =	64000						
max r-space proj	IRMAX =	1	max aug-charges	IRDMAX=	1821			
dimension x,y,z	NGX =	40	NGY =	40	NGZ =	40		
dimension x,y,z	NGXF=	80	NGYF=	80	NGZF=	80		
support grid	NGXF=	80	NGYF=	80	NGZF=	80		
ions per type =		1						
NGX,Y,Z	is equivalent	to a cutoff of	8.31,	8.31,	8.31	a.u.		
NGXF,Y,Z	is equivalent	to a cutoff of	16.62,	16.62,	16.62	a.u.		

SYSTEM = 0 atom

POSCAR = 3D\Atomistic\ (2)

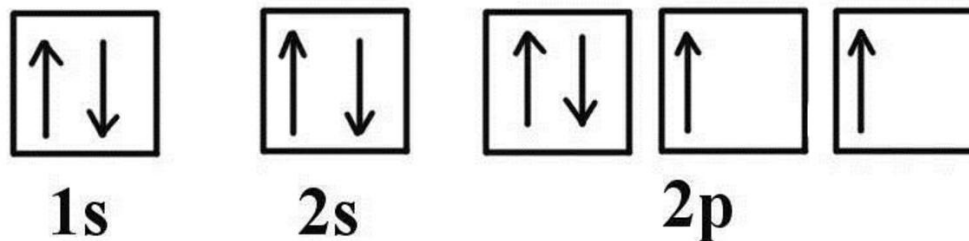
Startparameter for this run:

NWRITE =	2	write-flag & timer
PREC =	normal	normal or accurate (medium, high low for compatibility)
ISTART =	0	job : 0-new 1-cont 2-samecut
ICHARG =	2	charge: 1-file 2-atom 10-const
ISPIN =	1	spin polarized calculation?
LNONCOLLINEAR =	F	non collinear calculations
LSORBIT =	F	spin-orbit coupling
INIWAV =	1	electr: 0-lowe 1-rand 2-diag
LASPH =	F	aspherical Exc in radial PAW
METAGGA=	F	non-selfconsistent MetaGGA calc.

SCF迭代

```
----- Iteration      1(  1) -----  
  
POTLOK:  cpu time    0.0214: real time    0.0306  
SETDIJ:  cpu time    0.0058: real time    0.0069  
EDDAV:   cpu time    0.0979: real time    0.1333  
  DOS:   cpu time    0.0030: real time    0.0049  
-----  
  LOOP:  cpu time    0.1281: real time    0.1757  
  
eigenvalue-minimisations :   128  
total energy-change (2. order) : 0.2707589E+02  (-0.1531478E+03)  
number of electron        6.0000000 magnetization  
augmentation part        6.0000000 magnetization  
  
Free energy of the ion-electron system (eV)  
-----  
alpha Z      PSCENC =          0.27139542  
Ewald energy TEWEN =         -91.92708002  
-Hartree energ DENC =        -281.84385743  
-exchange    EXHF  =          0.00000000  
-V(xc)+E(xc) XCENC =          26.06853627  
PAW double counting =        346.54947689      -348.34814756  
entropy T*S   EENTRO =          0.00057979  
eigenvalues   EBANDS =         -55.95821233  
atomic energy EATOM =          432.26319604  
Solvation     Ediel_sol =          0.00000000  
-----  
free energy    TOTEN  =          27.07588707 eV  
  
energy without entropy =          27.07530728  energy(sigma->0) =          27.07569380  
-----
```

Oxygen



```

778 E-fermi : -8.9020      XC (G=0) : -0.8052      alpha+bet : -0.1463
779
780
781 k-point    1 :          0.0000      0.0000      0.0000
782 band No.   band energies      occupation
783     1      -23.8438          2.00000      2s
784     2       -8.9040          1.33333      2p
785     3       -8.9040          1.33333      2p
786     4       -8.9040          1.33333      2p
787     5       -0.4684          0.00000
788     6        1.8626          0.00000
789     7        1.8626          0.00000
790     8        1.8626          0.00000
791     9        1.9953          0.00000
792    10        2.0320          0.00000
  
```

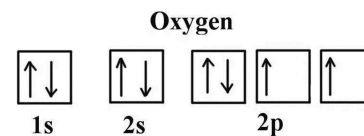
思考：为什么这里没有1s的能级？

由于氧原子具有磁性，**自旋极化**计算要进行考虑
在 VASP 计算中，需要在 INCAR 中添加一项：**ISPIN = 2**

```
1 SYSTEM = O
2 ISMAER = 0
3 SIGMA = 0.01
4 ISPIN = 2
```

```
[a2s000087@ln32%bscc-a2 O]$ cat OSZICAR
```

	N	E	dE	d eps	ncg	rms	rms (c)
DAV:	1	0.265523369297E+02	0.26552E+02	-0.13545E+03	256	0.355E+01	
DAV:	2	-0.106175931862E+01	-0.27614E+02	-0.27613E+02	384	0.141E+01	
DAV:	3	-0.111220265639E+01	-0.50443E-01	-0.49284E-01	256	0.869E-01	
DAV:	4	-0.111225447287E+01	-0.51816E-04	-0.51815E-04	512	0.310E-02	
DAV:	5	-0.111225447228E+01	0.59373E-09	-0.10125E-09	256	0.440E-05	0.654E+00
DAV:	6	0.311903474717E+00	0.14242E+01	-0.32471E+00	320	0.112E+00	0.149E+00
DAV:	7	-0.976724981987E+00	-0.12886E+01	-0.80277E-01	384	0.499E-01	0.540E-01
DAV:	8	-0.134013159441E+01	-0.36341E+00	-0.74679E-02	320	0.162E-01	0.240E-01
DAV:	9	-0.147860569084E+01	-0.13847E+00	-0.20730E-02	384	0.829E-02	0.131E-01
DAV:	10	-0.152073784222E+01	-0.42132E-01	-0.82524E-03	320	0.516E-02	0.231E-02
DAV:	11	-0.152204752794E+01	-0.13097E-02	-0.52040E-04	384	0.117E-02	0.859E-03
DAV:	12	-0.152238693311E+01	-0.33941E-03	-0.29099E-05	320	0.347E-03	0.377E-03
DAV:	13	-0.152241537065E+01	-0.28438E-04	-0.42957E-06	384	0.113E-03	
1	F= -.15224154E+01 E0= -.15211708E+01 d E =-.373359E-02 mag= 1.9998 磁矩: 2						



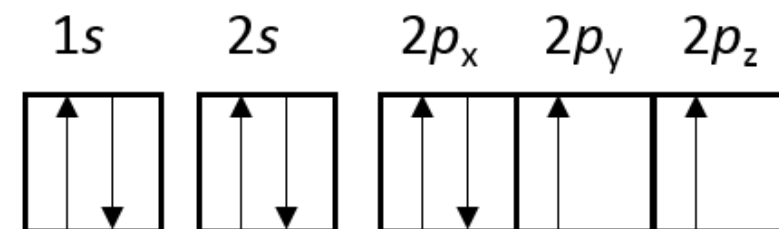
E-fermi : -7.0567 XC (G=0): -0.7802 alpha+bet : -0.1463

spin component 1

k-point	1 :	0.0000	0.0000	0.0000
band No.	band energies	occupation		
1	-25.0879	1.00000		
2	-10.0832	1.00000		
3	-10.0832	1.00000		
4	-10.0832	1.00000		
5	-0.5203	0.00000		
6	1.8066	0.00000		
7	1.8066	0.00000		
8	1.8066	0.00000		
9	1.8383	0.00000		
10	2.0200	0.00000		

spin component 2

k-point	1 :	0.0000	0.0000	0.0000
band No.	band energies	occupation		
1	-21.8402	1.00000		
2	-7.0547	0.33333		
3	-7.0547	0.33333		
4	-7.0547	0.33333		
5	-0.3359	0.00000		
6	2.0074	0.00000		
7	2.0074	0.00000		
8	2.0074	0.00000		
9	2.0267	0.00000		
10	2.1741	0.00000		



上面的结果是由于体系的高对称性导致的简并所造成的。
8*8*8 的立方体格子，在这里具有高阶的点群对称性：O_h

```

121 Analysis of symmetry for initial positions (statically):
122 =====
123 Subroutine PRICEL returns:
124 Original cell was already a primitive cell.
125
126
127 Routine SETGRP: Setting up the symmetry group for a
128 simple cubic supercell.
129
130
131 Subroutine GETGRP returns: Found 48 space group operations
132 (whereof 48 operations were pure point group operations)
133 out of a pool of 48 trial point group operations.
134
135
136 The static configuration has the point symmetry O_h .
137
138
139 Analysis of symmetry for dynamics (positions and initial velocities):
140 =====

```

在 GGA 泛函中，为了获得体系更低的能量，对于原子来说，通常会采用一种 symmetry broken solution 的处理方法。但是在 VASP 计算中，体系的对称性则是通过晶胞来获得，即把这个晶胞当成一个原子来处理，因此我们需要手动改变晶胞的形状来消除对称性造成的简并结果。

降低supercell的对称性

```
1 3D\Atomistic\ (2)
2 1.0
3      7.5000000000      0.0000000000      0.0000000000
4      0.0000000000      8.0000000000      0.0000000000
5      0.0000000000      0.0000000000      8.9000000000
6      0
7      1
8 Direct
9      0.500000000      0.500000000      0.500000000
```

INCAR中ISYM=0 直接关掉对称性

```
1 SYSTEM = 0
2 ISMAER = 0
3 SIGMA = 0.01
4 ISPIN = 2
5 ISYM = 0
```

```
[a2s000087@ln32%bscc-a2 0]$ cat OSZICAR
```

	N	E	dE	d eps	ncg	rms	rms (c)	
DAV:	1	0.275380726414E+02	0.27538E+02	-0.13356E+03	256	0.335E+01		
DAV:	2	-0.105860212164E+01	-0.28597E+02	-0.28596E+02	384	0.137E+01		
DAV:	3	-0.111178455264E+01	-0.53182E-01	-0.50600E-01	256	0.908E-01		
DAV:	4	-0.111185295400E+01	-0.68401E-04	-0.68402E-04	512	0.363E-02		
DAV:	5	-0.111185295418E+01	-0.18031E-09	-0.14941E-09	256	0.527E-05	0.654E+00	
DAV:	6	0.387610457657E+00	0.14995E+01	-0.33456E+00	448	0.114E+00	0.338E+00	
DAV:	7	-0.968580011539E+00	-0.13562E+01	-0.72854E-01	448	0.634E-01	0.536E+00	
DAV:	8	-0.102766125064E+01	-0.59081E-01	-0.13371E-02	256	0.145E-01	0.488E+00	
DAV:	9	-0.135798509052E+01	-0.33032E+00	-0.78166E-02	448	0.263E-01	0.352E+00	
DAV:	10	-0.182954298612E+01	-0.47156E+00	-0.52667E-01	448	0.713E-01	0.598E-01	
DAV:	11	-0.188088680262E+01	-0.51344E-01	-0.83701E-03	448	0.897E-02	0.220E-01	
DAV:	12	-0.188872416694E+01	-0.78374E-02	-0.19002E-03	448	0.376E-02	0.707E-02	
DAV:	13	-0.189018545917E+01	-0.14613E-02	-0.30340E-04	512	0.166E-02	0.596E-02	
DAV:	14	-0.189103883779E+01	-0.85338E-03	-0.22877E-04	384	0.155E-02	0.177E-02	
DAV:	15	-0.189142951747E+01	-0.39068E-03	-0.92519E-05	384	0.675E-03	0.140E-02	
DAV:	16	-0.189169195844E+01	-0.26244E-03	-0.45562E-05	448	0.527E-03	0.632E-03	
DAV:	17	-0.189214150028E+01	-0.44954E-03	-0.25508E-05	384	0.402E-03	0.122E-02	
DAV:	18	-0.189220558006E+01	-0.64080E-04	-0.11868E-05	384	0.343E-03		
1	F=	-.18922056E+01	E0=	-.18922056E+01	d E	=0.105816E-28	mag=	2.0001

OUTCAR中提示 降低对称性

```
IMPORTANT INFORMATION: All symmetrisations will be switched off!  
NOSYMM: (Re-)initialisation of all symmetry stuff for point group C_1.
```



```

----- Iteration 1 ( 18) -----

POTLOK:  cpu time    0.0170: real time    0.0247
SETDIJ:  cpu time    0.0025: real time    0.0025
EDDAV:   cpu time    0.1647: real time    0.1647
DOS:     cpu time    0.0002: real time    0.0002
-----
LOOP:    cpu time    0.1844: real time    0.1921

eigenvalue-minimisations : 384
total energy-change (2. order) :-0.6407979E-04 (-0.1186763E-05)
number of electron      5.9999968 magnetization      2.0000854
augmentation part       0.5207612 magnetization      0.4331097

Free energy of the ion-electron system (eV)
-----
alpha Z      PSCENC =          0.26021434
Ewald energy TEWEN =         -89.95113396
-Hartree energ DENC =        -285.40143342
-exchange    EXHF  =          0.00000000
-V(xc)+E(xc) XCENC =          26.94868410
PAW double counting =          360.09066278      -361.86776995
entropy T*S   EENTRO =          0.00000000
eigenvalues   EBANDS =         -84.23462549
atomic energy EATOM =          432.26319604
Solvation     Ediel_sol =          0.00000000
-----
free energy   TOTEN  =          -1.89220558 eV

energy without entropy =          -1.89220558  energy(sigma->0) =          -1.89220558

```

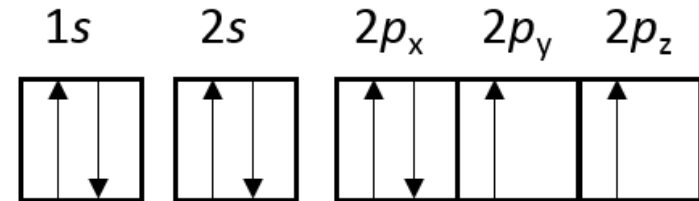
能量更低

E-fermi : -7.3482 XC (G=0) : -0.7350 alpha+bet : -0.1402

spin component 1

k-point 1 : 0.0000 0.0000 0.0000

band No.	band energies	occupation
1	-25.1089	1.00000
2	-10.7857	1.00000
3	-10.7836	1.00000
4	-8.6838	1.00000
5	-0.5274	0.00000
6	1.4381	0.00000
7	1.5790	0.00000
8	1.7666	0.00000
9	1.9663	0.00000
10	2.0590	0.00000



spin component 2

k-point 1 : 0.0000 0.0000 0.0000

band No.	band energies	occupation
1	-21.4438	1.00000
2	-7.4288	1.00000
3	-6.3233	0.00000
4	-6.3226	0.00000
5	-0.3035	0.00000
6	1.5199	0.00000
7	1.7802	0.00000
8	2.0961	0.00000
9	2.1078	0.00000
10	2.4001	0.00000

H₂ 原子计算: INCAR (写INCAR不要有Tab键)

在终端输入vi INCAR把以下的字符输进去, #号以后的起注释作用, 写不写都无所谓

- SYSTEM = H2 atom # H₂分子
- ISMEAR = 0 # 是零, 不是氧符号!
- SIGMA = 0.01 # 对于分子或者原子, 用0.01

➤ 每个参数后面跟着等号 = ,

➤ 等号前后可以有空格, 也可以没有, ISMEAR=0 和 ISMEAR = 0 结果一样,

➤ SYSTEM 后面是该计算的说明, 这个可以随便写, 本行也可以删除。

➤ ISMEAR这一项, 不同的值对应的是不同的展宽方法,

1. 对于半导体和绝缘体体系, ISMEAR的值**绝对不能大于0**, 一般用0,
2. 金属体系一般用1, 0亦可以

如果不知道用啥, ISMEAR=0,可以满足大部分的体系

➤ SIGMA该项,

1. 如果用了 ISMEAR = -5 , 可以忽略;
2. 对于金属: ISMEAR = 1 或者0, 非金属: ISMEAR = 0 的时候, 一般取0.10 或者采用默认值 0.20 即可。
3. 对于气体分子, 原子体系, ISMEAR = 0; SIGMA = 0.01 (标配)

SIGMA 的取值要保证OUTCAR 中的 entropy T*S 这一项, 平均到每个原子上小于 1meV。可使用grep 'entropy T' OUTCAR

H₂分子计算：KPOINTS (5行)

```
K-POINTS # 第一行随便写都行，不能没有  
0        # 零，格子自动生成  
Gamma    # gamma centered Monkhorst-Pack Grid  
1 1 1    # 1*1*1格子  
0 0 0    # S1 S2 S3, 一般保持 0 0 0 不变。
```

- 1) 第二行是数字0，不是字母O，0 在这里表示的是自动生成K点。
- 2) 第三行，VASP只认第一个字母，大小写均可。另外一种原始的Monkhorst-Pack 网格，二者的区别是M在G的基础上在三个方向平移了1/2N个单位，所以G点是M点的特殊情况
- 3) 第四行，在xyz三个方向上生成对应数目的K点，本例是1*1*1；
- 4) 第五行，一般都写成 0 0 0 ；
- 5) **对于六角晶系hexagonal的结构，用gamma centered!** 这一点，VASP官网说的很明确；
- 6) 气体分子，原子的计算用gamma

POTCAR

```
PAW_PBE H 15Jun2001
1.0000000000000000
parameters from PSCTR are:
VRHFIN =H: ultrasoft test
LEXCH = PE
EATOM = 12.4884 eV, 0.9179 Ry

TITEL = PAW_PBE H 15Jun2001
LULTRA = F use ultrasoft PP ?
IUNSCR = 0 unscreen: 0-lin 1-nonlin 2-no
RPACOR = 0.000 partial core radius
POMASS = 1.000; ZVAL = 1.000 mass and valenz
RCORE = 1.100 outmost cutoff radius
RWIGS = 0.700; RWIGS = 0.370 wigner-seitz radius (au A)
ENMAX = 250.000; ENMIN = 200.000 eV
RCLOC = 0.701 cutoff for local pot
LCOR = T correct aug charges
LPAW = T paw PP
EAUG = 400.000
RMAX = 1.123 core radius for proj-oper
RAUG = 1.200 factor for augmentation sphere
RDEP = 1.112 radius for radial grids
RDEPT = 0.926 core radius for aug-charge
```

体系中有多种元素时：使用cat命令

cat C H O N >POTCAR

切记和POSCAR中元素顺序保持一致!!!

- VRHFIN 用来看元素的价电子排布,
- LEXCH 表示这个POTCAR对应的是GGA-PBE泛函; 如果INCAR中不设定泛函, 则默认通过这个参数来设定。
- ZVAL 指的是实际上POTCAR中价电子的数目, 做Bader电荷分析的时候, 极其重要。
- ENMAX 代表默认的截断能。与INCAR中的ENCUT这个参数相关
- TITEL 指的是哪个元素, 以及POTCAR发布的时间

查看POTCAR有哪些元素 `grep TIT POTCAR`
以此类推

H₂分子计算：POSCAR，把H₂分子放在8x8x8的晶胞里

H₂ atom in a box #第一行随便写，不能没有

1.0 #缩放系数

8.0 0.0 0.0

0.0 8.0 0.0

0.0 0.0 8.0 #组成格子的坐标信息，从原点出发

H # H₂分子

2 #H原子的数目

cart #笛卡尔坐标cart，分数坐标direct

0.50 0.50 0.5

0.575600 0.5528299 0.507110 #H2分子位置

.cif结构文件除了用VESTA转换，也可以直接由vaspkit产生POSCAR，具体操作输入：vaspkit → 105

试着跑一跑

结构优化

通过移动原子位置，寻找能量最小，最稳定的结构

- 分子体系的优化: H_2 , O_2 , NH_3 , CH_4
- 晶体晶格常数的优化
- 表面的优化: 固定底层的原子
 - 简单金属表面，表面重构等
 $Au(111)$, $Au(997)$, $Au(110)$ (2×1) 重构
 - 氧化物表面，终止面 ZnO
 - 复合结构等，石墨烯和h-BN单层
 - 吸附体系
- Structure optimization
 - `IBRION = 1` RMM-DIIS
 - `IBRION = 2` conjugate gradient 一般IBRION取2
 - `IBRION = 3` damped molecular dynamics

```
SYSTEM=MgO-opt
ICHARG=2
LCHARG=.FALSE.
LWAVE=.FALSE.
LREAL=Auto
#SCF
ALGO=Normal
EDIFF=1E-5
ISMEAR=0
SIGMA=0.1
PREC=Normal
NELM=150
NELMIN=6
ENCUT=400
#OPT
EDIFFG=-0.02
NSW=0
IBRION=2 # 0: MD, 1: Newton, 2: CG
ISIF=2 # 3 for bulk, 2 for surface and
others
LORBIT=11
ISPIN=2
```

ISIF = 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8

Default: ISIF = 0 for IBRION = 0 (molecular dynamics) or LHFALC = .TRUE.
= 2 else

Description: Determines if the stress tensor is calculated and which ionic degrees of freedom are varied.

For ISIF ≥ 2, the stress tensor is calculated. It is defined as the negative of the derivative of the energy E with respect to the strain tensor η_{ji} :

$$\sigma_{ij} = -\frac{\delta E}{\delta \eta_{ji}}.$$

This might be different from other first principles codes. A positive in the diagonals means that the system is under compressive strain and wants to expand. A negative value implies that the system is under tensile strain and wants to reduce its volume. The stress tensor is symmetric $\sigma_{ij} = \sigma_{ji}$, and, thus, it has six independent entries. The calculation of the stress tensor is relatively time-consuming, and, therefore, by default, it is switched off in some cases. The forces are always calculated.

Tip: You can get information about the stress at each ionic step using NWRITE = 0, 1, 2, 3.

ISIF also determines which degrees of freedom (ionic positions, cell volume, and cell shape) of the structure are allowed to change.

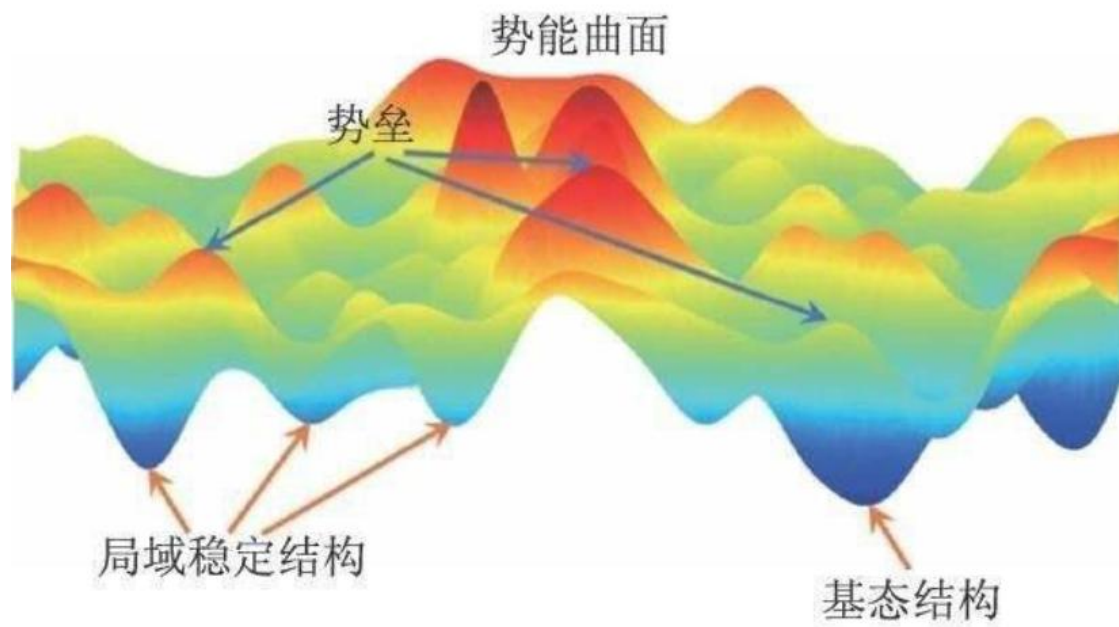
ISIF	calculate		degrees-of-freedom		
	forces	stress tensor	positions	cell shape	cell volume
0	yes	no	yes	no	no
1	yes	trace only	yes	no	no
2	yes	yes	yes	no	no
3	yes	yes	yes	yes	yes
4	yes	yes	yes	yes	no
5	yes	yes	no	yes	no
6	yes	yes	no	yes	yes
7	yes	yes	no	no	yes
8	yes	yes	yes	no	yes

规定是否计算力，应力张量；
是否优化原子位置，晶胞形状，晶胞大小

一般ISIF取2

分子和团簇的优化

1. 构建大的supercell，在考量计算量的情况下，保证近邻image中最近原子距离 $>6\text{\AA}$ （可以认为破坏对称性）
2. 只优化原子坐标，不优化晶格常数：VASP中ISIF=2



示例：H₂分子的结构优化，修改INCAR，POSCAR

INCAR

SYSTEM = H2
ISMAER = 0
SIGMA = 0.01
ENCUT = 400
NSW = 200
EDIFF = 1E-5
EDIFFG = -0.02
IBRION = 2
IVDW = 11
PREC = Normal
ISIF = 2

POSCAR

H2
1.0
9.0000000000 0.0000000000 0.0000000000
0.0000000000 10.0000000000 0.0000000000
0.0000000000 0.0000000000 9.5000000000
H
2
Direct
0.500000000 0.500000000 0.500000000
0.565970004 0.536059976 0.473140001

EDIFFG 结构优化收敛条件

[Page](#) [Discussion](#) [Read](#) [View source](#) [View history](#)

EDIFFG = [real]
Default: **EDIFFG** = **EDIFF** × 10

Description: **EDIFFG** defines the break condition for the ionic relaxation loop.

When **EDIFFG** is positive, the relaxation is stopped when the change of the total energy is smaller than **EDIFFG** between two ionic steps.

When **EDIFFG** is negative, the relaxation is stopped when the norms of all the forces are smaller than |**EDIFFG**|. This is usually a more convenient setting.

If **EDIFFG** = 0, the ionic relaxation is stopped after **NSW** steps.

计算结果能量提取

- 首先需判断体系有没有收敛，使用命令grep reached OUTCAR

```
[a2s000087@ln32%bscc-a2 h2-opt]$ grep reached OUTCAR
```

```
----- aborting loop because EDIFF is reached -----  
----- aborting loop because EDIFF is reached -----  
----- aborting loop because EDIFF is reached -----  
reached required accuracy - stopping structural energy minimisation
```

SCF收敛

离子步收敛

- tail -1 OSZICAR 或grep E0 OSZICAR提取最终能量，取最后一个E0的能量：

```
[a2s000087@ln32%bscc-a2 h2-opt]$ grep E0 OSZICAR
```

```
1 F= -.67565285E+01 E0= -.67567218E+01 d E =-.675653E+01  
2 F= -.67476208E+01 E0= -.67478140E+01 d E =0.890775E-02  
3 F= -.67585762E+01 E0= -.67587694E+01 d E =-.204766E-02
```

晶格常数优化

- 晶格常数优化的两种方法：
 - 程序自动，VASP中INCAR中设置ISIF = 3
 - 手动法，保持晶格参数比例，对晶胞进行缩放，得到能量与缩放系数之间的关系图，取能量的最低点
- 实例：用两种方法优化Si的晶格常数
- exp: $a = 3.84$

思考：为什么从实验得来的晶体结构需要作结构优化？

自动优化法:

```
INCAR
SYSTEM = Si;  PREC  = Normal;  ENCUT = 500 eV ;
EDIFF = 1E-5;  EDIFFG = -0.02;  LREAL = F
LCHARG = F;  LWAVE = F;  GGA = PE
NSW = 300;  IBRION = 2;  ISIF = 3;
```

Si-bulk

1.0

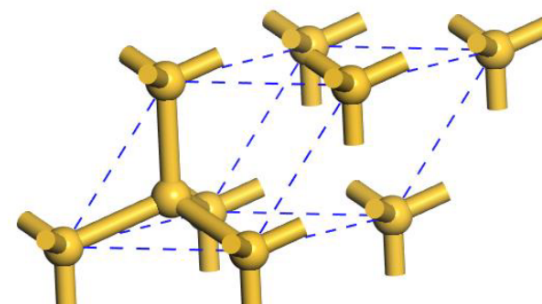
3.8401000500	0.0000000000	0.0000000000
1.9200500250	3.3256241964	0.0000000000
1.9200500250	1.1085413988	3.1354285612

Si

2

Direct

0.0000000000	0.0000000000	0.0000000000
0.2500000000	0.2500000000	0.2500000000



KPOINTS:

Auto

0

G

9 9 9

0. 0. 0.

存储优化后的结构

CONTCAR

Si-bulk

1.0000000000000000

3.8668082709276259 0.0000000000000000 0.0000000000000000

1.9334041354638130 3.3487541942137558 -0.0000000000000000

1.9334041354638130 1.1162513980712518 3.1572357322691214

Si

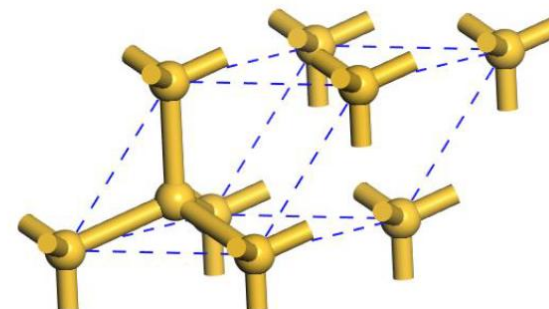
2

Direct

0.0000000000000000 -0.0000000000000000 -0.0000000000000000

0.2500000000000000 0.2500000000000000 0.2500000000000000

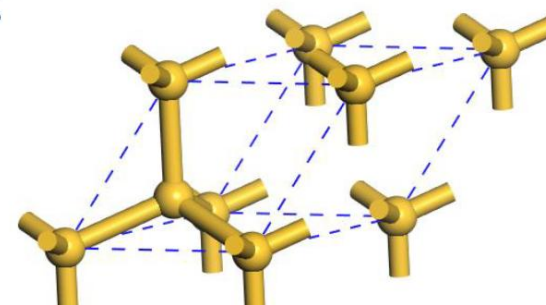
晶格常数由3.8401变为3.8668
略微增大，属于正常情况



手动法，搜索极小值：

INCAR

SYSTEM = Si; PREC = Normal; ENCUT = 500 eV ;
EDIFF = 1E-5; EDIFFG = -0.02; LREAL = F
LCHARG = F; LWAVE = F; GGA = PE
NSW = 300; IBRION = 2; **ISIF = 2;**



Si-bulk

0.95-1.06 (用for循环控制)

3.8401000500	0.0000000000	0.0000000000
1.9200500250	3.3256241964	0.0000000000
1.9200500250	1.1085413988	3.1354285612
Si		
2		
Direct		
0.0000000000	0.0000000000	0.0000000000
0.2500000000	0.2500000000	0.2500000000

KPOINTS:

Auto

0

G

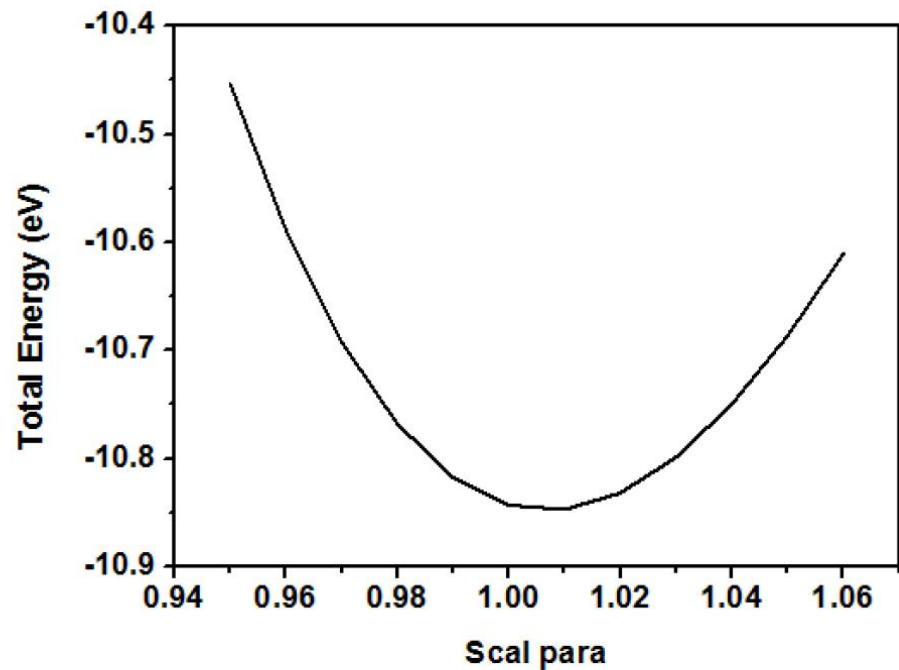
9 9 9

0. 0. 0.


```

for i in 0.95 0.96 0.97 0.98 0.99 1.00 1.01 1.02 1.03 1.04 1.05 1.06
do
cat > POSCAR <<!
Si-bulk
$i
      3.8401000500      0.0000000000      0.0000000000
      1.9200500250      3.3256241964      0.0000000000
      1.9200500250      1.1085413988      3.1354285612
      Si
      2
Direct
      0.0000000000      0.0000000000      0.0000000000
      0.2500000000      0.2500000000      0.2500000000
!
mpirun -np $1 $EXEC > log
E=`grep "without entropy" OUTCAR |tail -n 1|awk '{printf $7}'`
cputime=`grep "CPU" OUTCAR |awk '{printf $6}'`
echo $i $E $cputime >> comment
mv CONTCAR CONTCAR-$i
mv OUTCAR OUTCAR-$i
done

```

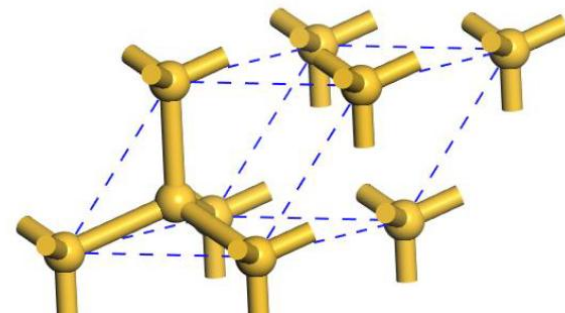


Scale=1.01

得到晶胞参数

$a=b=c = 3.8401 * 1.01 \text{ \AA} = 3.878501 \text{ \AA}$

比实验值大1.000%



表面优化

1. 固定底层的原子以模拟bulk环境，通常不优化晶格常数
2. 优化表面结构，优化吸附物结构，优化吸附体系结构（不同的吸附位置）
3. 计算吸附能 $E_{\text{ads}} = E_{\text{M/surf}} - (E_{\text{M}} + E_{\text{surf}})$

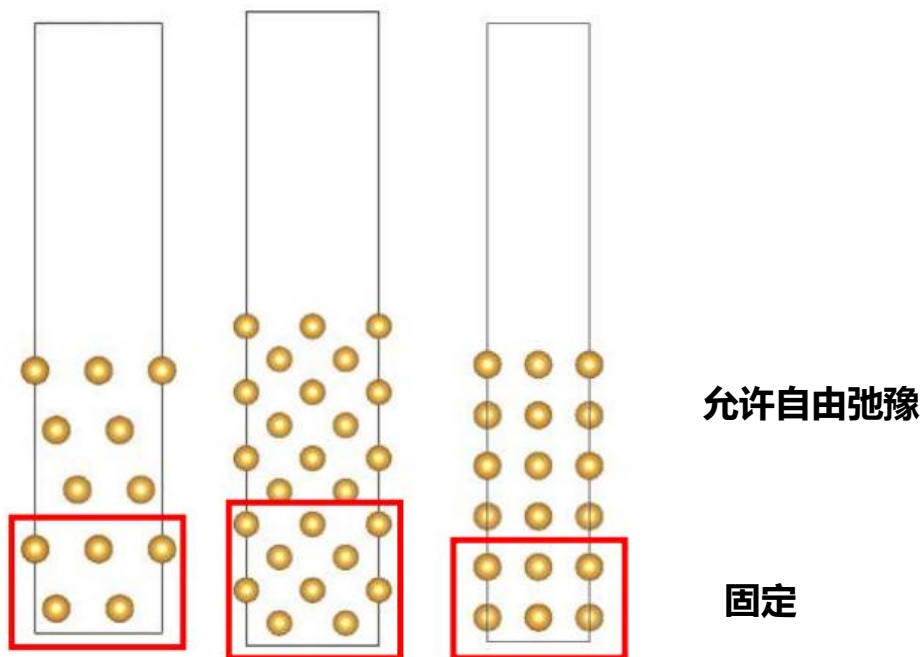
不同文献中可能会给出相反的定义，注意吸附能正负表示的意义。

```
Cu-CHO
1.0
      5.1119999886      0.0000000000      0.0000000000
      -0.0000088492      5.1119999885      0.0000000000
      -0.0000266965     -0.0000266965     15.4221000671
      Cu      C      O      H
      16      1      1      1
Selective dynamics
Direct
      0.000030000      0.001330000      0.179790005      F      F      F
      0.000100000      0.005870000      0.410259992      T      T      T
      0.250000000      0.250000000      0.064839996      F      F      F
      0.255829990      0.251619995      0.299970001      T      T      T
      0.500010014      0.000420000      0.179309994      F      F      F
      0.500079989      0.003300000      0.420670003      T      T      T
      0.750000000      0.250000000      0.064839996      F      F      F
      0.744279981      0.251610011      0.299959987      T      T      T
      0.000020000      0.499980003      0.180910006      F      F      F
      0.000020000      0.502420008      0.413540006      T      T      T
      0.250000000      0.750000000      0.064839996      F      F      F
      0.253170013      0.752380013      0.299329996      T      T      T
      0.500020027      0.500800014      0.179820001      F      F      F
      0.500029981      0.493849993      0.421799988      T      T      T
      0.750000000      0.750000000      0.064839996      F      F      F
      0.746959984      0.752409995      0.299309999      T      T      T
      0.500989974      0.632679999      0.538280010      T      T      T
```

POSCAR坐标固定

默认结构优化都是允许所有原子在所有的方向上(x, y, z)做自由运动，有些情况下结构优化计算需要固定部分原子。

比如：表面计算需要固定下层原子，固定slab模型的经验规律一般都是下两层，或者三层。一般让可弛豫的**原子等于或多于被固定的层数**



VASP部分固定原子坐标方法 (POSCAR)

- (1) 在POSCAR第7行和第8行之间插入一行， Selective Dynamics 前面讲过VASP只认第一个字母， 也就是S是必须的。
- (2) 加入Selective之后， 需要在每一行原子坐标的后面加上T或者F表示允许和禁止移动。这里我们需要加三个T或者F， 表示在x y z 三个方向上选择性固定原子的移动

```
optau\ (1\1\1)
1.0
      5.8734002113      0.0000000000      0.0000000000
     -2.9367001057      5.0865137896      0.0000000000
      0.0000000000      0.0000000000     24.5911998749
      Au
      20
  Selective Dynamics
  Direct
      0.333330005      0.666670024      0.040660001      F F F
      0.000000000      0.000000000      0.138170004      F F F
```

可以用 vaspkit 快速固定原子坐标

以Au(100) 为例，固定原子坐标

MS建模练习

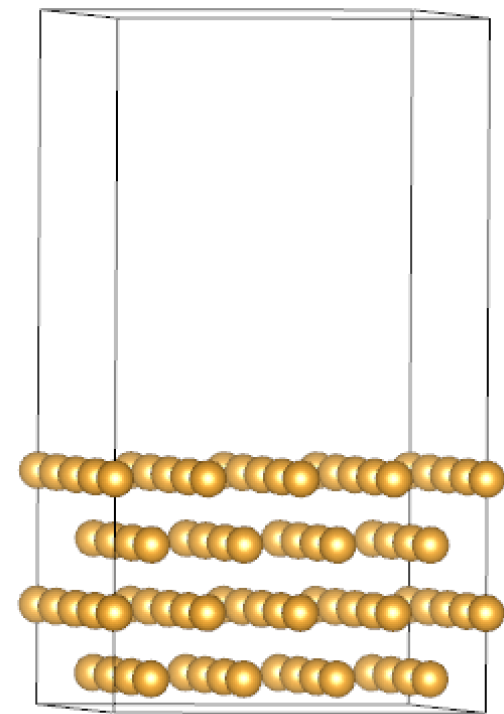
MS中建一个Au(100)表面。从MS自带结构库中导入，切表面时选择切2层，加15埃真空层，扩2*2*1的超胞，导出为cif文件，在VESTA中打开转换为VASP的输入文件POSCAR

vaspkit环境变量添加

在家目录的.bashrc中添加下面这行：

```
export PATH=/public3/home/a2s000087/software/vaspkit.1.4.1/bin:${PATH}
```

步骤： vim ~/.bashrc，粘贴这一句，然后:wq保存，然后source ~/.bashrc



vaspkit固定原子层数→402

方法一：根据原子层数固定

vaspkit可以自动的判断原子层数：
比如含有4层的Au(100)表面

流程：输入vaspkit → 402 → 因为是POSCAR,所以选
1 → Threshold是vaspkit的总层数判断标准，在不
同的标准下层数会判断的不太一样，选择
Threshold的数值,这里选择1，判断出来4层 → 我们
固定2层，所以输入2 → 选择固定方向，输入all →
检查一下生成的 POSCAR_FIX.vasp 文件，如果没
有问题： cp POSCAR_FIX.vasp POSCAR

此方法只限于固定z方向的slab模型

```
----->>
402
===== Structural File =====
1) POSCAR
2) CONTCAR
3) Specified Structure File

0) Quit
9) Back
----->>
1
-->> (01) Reading Structure from POSCAR File...
===== Fixed Modes =====
1) Fix Atoms by Atomic Indexes/Labels
2) Fix Atoms by Layers
3) Fix Atoms by Heights
4) Fix Atoms by SELECTED_ATOMS_LIST File

0) Quit
9) Back
----->>
2
Threshold: 0.02, Found layer number: 4
Threshold: 0.04, Found layer number: 4
Threshold: 0.06, Found layer number: 4
Threshold: 0.08, Found layer number: 4
Threshold: 0.10, Found layer number: 2
Please input the threshold value to separate layer

----->>
0.04
Found 4 layers in c direction.
Which layers do you want to Fix (From bottom to top)? (1<= & <=4)
(Free Format is Compatible, e.g., 1 2 1-3 all)

----->>
2
-->> (02) Written SELECTED_ATOMS_LIST File!
Input the direction index to fix [1 <= & <= 3]
1) x direction
2) y direction
3) z direction
(Free Format is Compatible, e.g., 1 2 1-3 all)

----->>
all
-->> (03) Written POSCAR_FIX.vasp File!
```

根据原子z坐标高度固定层数

方法二： vaspkit 固定原子402

首先要弄清楚我们需要固定哪些原子，和这些原子的z坐标，比如 Au(100)固定下两层原子。
找到我们要固定的原子中z坐标最大及最小的原子的坐标

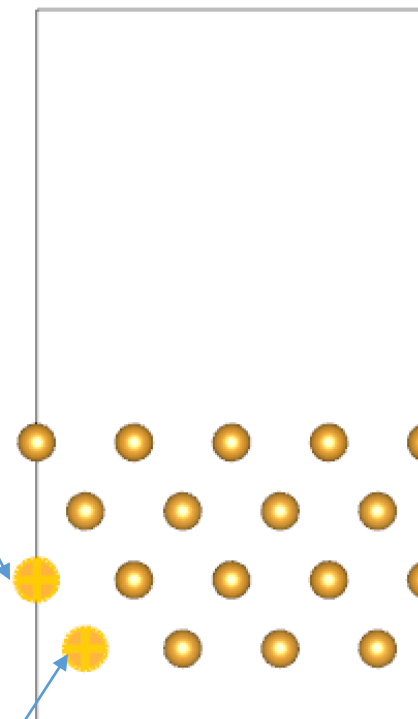
```
----->>
402
===== Structural File =====
1) POSCAR
2) CONTCAR
3) Specified Structure File

0) Quit
9) Back
----->>
1
-->> (01) Reading Structure from POSCAR File...
===== Fixed Modes =====
1) Fix Atoms by Atomic Indexes/Labels
2) Fix Atoms by Layers
3) Fix Atoms by Heights
4) Fix Atoms by SELECTED_ATOMS_LIST File

0) Quit
9) Back
----->>
3
Atoms between your chosen section in z direction will be fixed
Type z_min z_max, Note: z_min < z_max
----->>
0.102 0.2
Read and Write atomic coordinates as:
1) Fractional coordinates
2) Cartesian coordinates
----->>
1
-->> (02) Written SELECTED_ATOMS_LIST File!
Input the direction index to fix [1 <= & <= 3]
1) x direction
2) y direction
3) z direction
(Free Format is Compatible, e.g., 1 2 1-3 all)

----->>
all
-->> (03) Written POSCAR_FIX.vasp File!
```

这个原子的z坐标是0.19875,



Atom: 1	Au1 Au	1.00000	0.00000	0.19875	(1, 0, 0)+ x, y, z
	Occ. = 1.000		Ueq = 1.00000		1a 1
Atom: 15	Au15 Au	0.87500	0.12500	0.10219	(0, 0, 0)+ x, y, z
	Occ. = 1.000		Ueq = 1.00000		1a 1

这个原子的z坐标是0.10219, 所以我们只需要固定住所有z坐标在这两个坐标区间的原子就行了

随堂作业

NH₃: 平面分子, 三个NH键夹角120°, 初始键长设为1.2埃

- 用不同大小的基组优化H₂、O₂ (注意自旋态)、NH₃分子, 记录键长和能量的变化, 做出分析
- INCAR
 - ENCUT
 - IBRION = 2 (or 1)
 - ISMEAR = 0

课后作业：优化Cu的吸附

测试金属Cu的k点收敛情况以及截断能收敛情况，找到适合参数。可从结构网站或者MS内导入体相Cu的结构。

优化金属Cu的晶格常数（VASP），要求使用自动和手动两种方法

- 自动：ISIF=3 OSZICAR, OUTCAR, **CONTCAR**
- 手动扫描：改变不同的晶格参数，做单点能量计算
- 测试不同的泛函PBE、RPBE、vdw修正...（至少三个）(修改GGA参数)
- 搭建Cu(111)、Cu(100)表面，取不同层数，固定最下面两层，看层间距弛豫情况（用最小的原胞，真空层取10Å，注意原胞的k点密度需重新考虑）

作业要求：放上你的建模结构，写出你对参数设置的考虑，以及结果分析

遇到问题和报错怎么解决？

- VASP wiki: https://www.vasp.at/wiki/The_VASP_Manual
- VASP教程: <https://www.bigbrosci.com/zh-cn/latest/> 强推
- 计算化学公社: <http://bbs.keinsci.com/forum.php>
- 问助教

关于作业提交的要求

1. KSSOLV/VASP教学部分，预计安排3次随堂作业，4次课后作业；
2. **随堂作业**在当天下午课程结束前发助教邮箱，注意有时不止一道题，尽量存好草稿到第三堂课发送☺，文件名按群里规定，**txt格式**。满分100，只要交了至少会得75（学号+姓名，相当于签到）；在周二中午12点前提交，满分85，至少得60。逾期不受理；
3. **课后作业**截止时间为布置后下一周上课之前，格式为**word文档**，命名也是按照群里规定，要求有必要的图片和文字说明；
4. 今天的**随堂作业**发送到陆鹏助教邮箱：pluustc@mail.ustc.edu.cn
今天的**课后作业**发送到银栖麟助教邮箱：yxl170@mail.ustc.edu.cn

No_10_27Homework4_Name_StudentID.txt

No_10_27Homework3_Name_StudentID.docx

No改成你的序号